

CIVIL GENERAL NOTES:

1. REFER TO THE PROJECT SPECIFICATIONS IN ALL SITUATIONS. BRING CONFLICTS BETWEEN PROJECT SPECIFICATIONS AND PLANS TO THE ATTENTION OF THE CONTRACTING OFFICER AND/OR ENGINEER FOR RESOLUTION. ALL MATERIALS AND WORKMANSHIP MUST CONFORM TO THE PROJECT SPECIFICATIONS.

2. EXISTING UTILITY INFORMATION IS NOT TO BE RELIED ON AS BEING EXACT OR COMPLETE. LOCATION, VERIFICATION, SUPPORT AND PROTECTION OF ALL ABOVE GROUND AND UNDERGROUND UTILITIES TO REMAIN IN PLACE, INCLUDING ANY UTILITIES NOT INDICATED ON CONTRACT DRAWINGS IS REQUIRED. ANY CONFLICTS OR DISCREPANCIES MUST BE BROUGHT TO THE CONTRACTING OFFICER'S AND/OR ENGINEER'S ATTENTION IMMEDIATELY. CAPE FEAR ENGINEERING HAS NOT PHYSICALLY LOCATED ANY UNDERGROUND UTILITIES.

3. PRIOR TO ANY EXCAVATION, CONTRACTOR MUST HAVE THE EXISTING UNDERGROUND LINES MARKED ON THE GROUND SURFACE IN THE AREAS OF CONSTRUCTION. POTHOLE AND PHYSICALLY VERIFY DEPTH AND LOCATION OF ALL UTILITY LINES IN THE WORK AREA THAT REQUIRE COORDINATION WITH WORK ITEMS. POTHOLE VERIFICATION MUST BE DONE WELL IN ADVANCE OF ACTUAL WORK SO THAT POTENTIAL UTILITY CONFLICTS CAN BE VERIFIED AND RESOLVED WITHOUT CAUSING SCHEDULE DELAYS.

4. PROTECT ALL EXISTING STRUCTURES, TREES, PAVEMENT, UTILITIES, AND OTHER PROPERTY UNLESS THEY ARE TO BE DEMOLISHED. ANY PROPERTY NOT AUTHORIZED FOR REMOVAL, BUT DAMAGED BY THE CONTRACTOR MUST BE RESTORED BY THE CONTRACTOR TO THEIR PRE-CONSTRUCTION CONDITION OR BETTER AT NO ADDITIONAL EXPENSE TO THE OWNER.

5. MAKE EVERY EFFORT TO SAVE PROPERTY IRONS, MONUMENTS, OTHER PERMANENT POINTS AND LINES OF REFERENCE AND CONSTRUCTION STAKES. PROPERTY IRONS, MONUMENTS, AND OTHER PERMANENT POINTS OF REFERENCE DESTROYED BY THE CONTRACTOR MUST BE REPLACED AT THE CONTRACTOR'S EXPENSE.

6. WORK PERFORMED UNDER THIS CONTRACT MUST NOT IMPACT THE OPERATION OF ANY ADJACENT PROPERTIES UNLESS A UTILITY OUTAGE HAS BEEN APPROVED IN ADVANCE.

7. SEQUENCE WORK AS NECESSARY TO ENSURE THAT ALL UTILITY SERVICES, INCLUDING FIRE HYDRANTS, REMAIN OPERATIONAL DURING CONSTRUCTION.

8. RESTORE ANY ACCESS ROADS AND STAGING AREAS USED BACK TO THEIR PRE-CONSTRUCTION CONDITION OR BETTER. VEGETATE ALL DISTURBED AREAS IN ACCORDANCE WITH THE PROJECT SPECIFICATIONS.

9. TAKE ADEQUATE PROTECTIVE MEASURES TO ENSURE THAT EXISTING PAVEMENT SURFACES ARE NOT DAMAGED FROM TRACKED CONSTRUCTION EQUIPMENT OR ANY OTHER POSSIBLE SOURCE WHEN EQUIPMENT IS BEING MOVED. REPAIR ANY PAVED AREAS DAMAGED DURING CONSTRUCTION TO THEIR PRE-CONSTRUCTION CONDITION OR BETTER.

10. EXISTING PAVEMENTS AND ALL OTHER AREAS ADJACENT TO THE SITE MUST BE KEPT CLEAN AT ALL TIMES. DO NOT ALLOW BUILD UP OF SOIL, MUD, DUST OR OTHER DEBRIS.

11. CONTRACTOR IS RESPONSIBLE FOR ALL SURVEYING AND STAKING TO COMPLETE THE WORK.

12. ANY AND ALL MATERIAL QUANTITIES INDICATED ON THIS PLAN SET ARE FOR REFERENCE ONLY. CONTRACTOR MUST VERIFY ALL MATERIAL QUANTITIES SHOWN.

13. KEEP REDLINE MARK-UPS OF ANY CHANGES MADE FROM THE APPROVED PLAN AND PROVIDE COPY OF REDLINES TO ENGINEER UPON PROJECT COMPLETION.

14. THERE ARE POTENTIAL JURISDICTIONAL WETLANDS IN THE VICINITY OF THE PROJECT AREA. A MINIMUM 50-FT BUFFER SHALL BE MAINTAINED.

EXISTING CONDITION NOTES:

1. SEE SHEETS VF100 AND VF101 FOR EXISTING CONDITIONS AND SURVEY INFORMATION.

OFFSITE BORROW AREA NOTES

1. CONTRACTOR IS RESPONSIBLE FOR ENSURING OFFSITE BORROW AREA HAS AN APPROVED EROSION CONTROL PERMIT AND APPROPRIATE MEASURES IN PLACE PRIOR TO LAND DISTURBING ACTIVITY.

2. NO BORROW MATERIALS ARE AVAILABLE ON GOVERNMENT PROPERTY.

DEMOLITION PLAN NOTES:

1. LOCATE UNDERGROUND UTILITIES AND COORDINATE DEMOLITION AND / OR RELOCATION WORK. SPECIFICALLY TAKE EXTRA CAUTIONARY MEASURES TO PREVENT DISRUPTION OF ANY TELECOM FIBER IN THE PROJECT AREA. ONCE LOCATED IN THE FIELD, POSITIVE VERIFICATION BY POTHOLING IS REQUIRED. REPORT ANY CONFLICTS OR DISCREPANCIES TO ENGINEER.

2. LIMITS OF DEMOLITION AND DIMENSIONS ARE SHOWN FOR GENERAL REFERENCE. CONTRACTOR MUST BE RESPONSIBLE FOR REMOVAL OF ALL ITEMS, INCLUDING ALL EXCAVATED MATERIALS NECESSARY TO COMPLETE WORK WITHIN THIS CONTRACT.

3. WHERE ASPHALT/CONCRETE DEMOLITION IS REQUIRED, USE A SAW CUT AT THE LIMITS OF DEMOLITION OR NEAREST JOINT TO OBTAIN A CLEAN VERTICAL EDGE.

4. DEDICATED DEMOLITION AND OTHER WASTE AREAS/EARTHEN MATERIAL STOCKPILES MUST BE LOCATED AT LEAST 50' FROM STORMDRAINS OR STREAMS.

5. REPORT ANY UNSUITABLE SOILS, CONTAMINATED SOILS, AND BURIED CONCRETE AND DEBRIS TO THE CONTRACTING OFFICER IMMEDIATELY.

6. CONTRACTOR MUST PROVIDE TEMPORARY THRUST BLOCKS AND TEMPORARILY CAP WATERLINES FOR SEQUENCING OF WATERLINE DEMOLITION AND INSTALLATION OF NEW TEES AND FITTINGS.

7. CONTRACTOR MUST SEQUENCE DEMOLITION OF STORM AND SANITARY SEWER TO ENSURE ALL SYSTEMS REMAIN ONLINE DURING WORK. CONTRACTOR MUST PROVIDE TEMPORARY CAPS AND PLUGS AND BYPASS PUMPING AS REQUIRED FOR SEQUENCING OF DEMOLITION AND NEW WORK.

8. UNDERGROUND UTILITIES SHOWN ON THE DEMOLITION PLANS MAY EXTEND FURTHER THAN SHOWN. CONTRACTOR IS RESPONSIBLE FOR COMPLETE REMOVAL OF UTILITIES WITHIN THE ENTIRE PROJECT SITE UNLESS OTHERWISE NOTED.

SIGNAGE, STRIPING, AND MARKING NOTES:

1. THE CONTRACTOR IS RESPONSIBLE FOR ALL PEDESTRIAN AND VEHICULAR TRAFFIC DETOURS AND MUST USE APPROPRIATE SIGNAGE AND BARRICADES IN ACCORDANCE WITH ALL BASE, STATE, AND FEDERAL REQUIREMENTS.

2. ALL ROAD AND LANE CLOSURES MUST BE REQUESTED AND APPROVED BY THE CONTRACTING OFFICER PRIOR TO CLOSING OF ANY ROADWAYS.

3. ALL ROAD CLOSURES MUST BE SUBMITTED TO THE CONTRACTING OFFICER A MINIMUM OF 15 DAYS PRIOR TO THE REQUIRED DATE OF CLOSURE AND MUST INCLUDE A TRAFFIC CONTROL PLAN IN ACCORDANCE WITH MUTCD AND EM 385-1-1.

4. ALL TRAFFIC CONTROL METHODS AND DEVICES, BOTH TEMPORARY AND PERMANENT, MUST BE IN ACCORDANCE WITH THE U.S. DEPARTMENT OF TRANSPORTATION FEDERAL HIGHWAY ADMINISTRATION'S "MANUAL ON UNIFORM TRAFFIC CONTROL DEVICES" (MUTCD), MOST RECENT EDITION AND THE NORTH CAROLINA DEPARTMENT OF TRANSPORTATION (NCDOT).

5. USE THE COLOR WHITE FOR ALL PAVEMENT MARKINGS AND SYMBOLS UNLESS OTHERWISE NOTED.

6. DO NOT LOCATE PAVEMENT MARKING SYMBOLS AS TO ENCROACH INTO INTERSECTION AREAS.

7. DO NOT PLACE PAVEMENT MARKING SYMBOLS ACROSS TRANSVERSE EXPANSION JOINTS ON PORTLAND CONCRETE PAVEMENTS UNLESS APPROVED BY THE ENGINEER.

8. CONFORM ALL SYMBOLS TO THE NCDOT STANDARDS AND PROJECT SPECIFICATIONS AND DETAILS.

9. SEE SIGNAGE & PAVEMENT MARKINGS ON SITE PLAN AND DETAIL SHEETS, WHERE SHOWN.

10. PAVEMENT MARKINGS MUST BE COLORED PER THE PLANS. WHERE SHOWN, PAVEMENT MARKINGS LOCATED IN PARKING AREAS MUST BE WATER-BASED. CROSSWALKS AND PAVEMENT MARKINGS IN TRAFFIC AREAS MUST BE THERMOPLASTIC.

11. ANY EXISTING PAVEMENT MARKINGS IN THE VICINITY OF THE PROJECT LIMITS THAT ARE DAMAGED DURING CONSTRUCTION MUST BE REPAIRED OR REPLACED TO THE CONTRACTING OFFICER'S SATISFACTION.

SIDEWALK AND PAVING NOTES:

1. COMPLY WITH THE CURRENT NCDOT STANDARDS AND SPECIFICATIONS FOR ROADS AND STRUCTURES UNLESS OTHERWISE NOTED.

2. INSTALL EXPANSION JOINT WITH THICKENED EDGE WHERE CONCRETE PAVEMENT ABUTS BUILDINGS, MANHOLES, INLETS, AND OTHER STRUCTURES IN THE PAVEMENT UNLESS NOTED OTHERWISE.

3. FREE CONCRETE PAVEMENT EDGES THAT ABUT GRAVEL OR ASPHALT PAVEMENT AREAS MUST BE THICKENED.

4. CONSTRUCTION JOINTS MUST BE USED AT THE END OF EACH DAY'S PLACING OPERATIONS AND WHEN CONCRETE PLACEMENT IS INTERRUPTED FOR 30 MINUTES OR LONGER.

5. ODD SHAPED SLABS (WHERE L:W RATIO IS GREATER THAN 1.25) AND SLABS WITH PENETRATIONS (FOR UTILITIES, BASINS, BOLLARDS, ETC) MUST BE REINFORCED WITH WWF (6X6-W2.1X2.1) IN CENTER OF SLAB.

6. REFER TO PROJECT SPECIFICATIONS FOR SPACING AND DIMENSIONS OF ALL CONSTRUCTION AND EXPANSION JOINTS.

STORMWATER AND GRADING NOTES:

1. NO INSTALLATION OF IMPERVIOUS SURFACES IN EXCESS OF 10,000 SF MUST BE DONE PRIOR TO OBTAINING APPROVAL FROM NORTH CAROLINA DEPARTMENT OF ENVIRONMENTAL QUALITY (NCDEQ).

2. ALL CUT OR FILL SLOPES MUST BE 3:1 OR FLATTER UNLESS OTHERWISE NOTED.

3. DURING CONSTRUCTION, PERFORM GRADING IN A MANNER AND SEQUENCE THAT WILL PROVIDE PROPER DRAINAGE AT ALL TIMES.

4. ADJUST AND/OR CUT EXISTING PAVEMENT AS NECESSARY TO ASSURE A SMOOTH FIT AND CONTINUOUS GRADE.

5. FOR ALL NEW BUILDINGS, PROVIDE A MINIMUM 5% SLOPE AWAY FROM THE BUILDING FOR THE FIRST 10-FEET AND ASSURE THAT FINISHED FLOOR ELEVATIONS ARE A MINIMUM OF 6-INCHES ABOVE THE ADJACENT SITE GRADE IN NON-PAVED AREAS.

6. ALL STORM PIPE MUST BE CLASS III REINFORCED CONCRETE PIPE (RCP), UNLESS OTHERWISE NOTED ON THE PLANS.

7. UPON THE COMPLETION OF CONSTRUCTION AND SITE STABILIZATION, ALL NEW PIPES AND EXISTING PIPES IMPACTED BY CONSTRUCTION MUST BE CLEANED TO REMOVE ALL ACCUMULATED SILT, SEDIMENT AND DEBRIS.

8. ALL STORM PIPE ENTERING STRUCTURES (NEW AND EXISTING) MUST BE CUT FLUSH TO INSIDE OF THE BOX AND GROUTED TO ASSURE CONNECTION AT STRUCTURE IS WATER TIGHT; MAXIMUM PROTRUSION MUST BE 6 INCHES.

9. ALL STORMWATER STRUCTURES IN PAVED AREAS MUST BE FLUSH WITH PAVEMENT AND BE HEAVY-DUTY TRAFFIC RATED.

GENERAL UTILITY NOTES:

1. ALL UTILITY MATERIALS, CONSTRUCTION, TESTING, AND WORKMANSHIP MUST CONFORM TO THE PROJECT SPECIFICATIONS, UFC STANDARDS, AND MUST MEET OR EXCEED NORTH CAROLINA DEPARTMENT OF ENVIRONMENTAL QUALITY AND CAMP LEJEUNE STANDARDS AND SPECIFICATIONS.

2. A NORTH CAROLINA LICENSED UTILITY CONTRACTOR MUST PERFORM ALL UTILITY WORK INVOLVED IN CONSTRUCTING THIS PROJECT.

3. COORDINATE WITH THE UTILITY PROVIDERS FOR REQUIRED UTILITY ADJUSTMENTS AND/OR RELOCATIONS.

4. ISOLATE ALL NEW PIPING FROM EXISTING PIPING UNTIL ALL NEW PIPING HAS BEEN TESTED, CERTIFIED TO UTILITY PROVIDER AND ACCEPTED FOR SERVICE.

5. FIELD VERIFY THE LOCATION AND DEPTH OF ALL EXISTING UTILITY LINES PRIOR TO THE INSTALLATION OF ANY NEW LINES.

6. ALL FILL MATERIAL IS TO BE IN PLACE, AND COMPACTED BEFORE INSTALLATION OF UTILITIES. IF NOT, CONTRACTOR MUST ENSURE THAT MINIMUM COVER REQUIREMENTS ARE MET AND THAT UTILITIES ARE ADEQUATELY PROTECTED.

7. ADJUST ALL UTILITY STRUCTURE RIM ELEVATIONS AS NECESSARY TO MATCH FINAL ELEVATIONS.

8. NOTIFY THE UTILITY PROVIDER'S INSPECTORS AND CONTRACTING OFFICER, AND ENGINEER AT LEAST 72 HOURS BEFORE CONNECTING TO ANY EXISTING LINE AND/OR BEGINNING UTILITY LINE INSTALLATION.

9. NOTIFY AND RECEIVE APPROVAL FROM THE UTILITY PROVIDER'S INSPECTORS AND CONTRACTING OFFICER AT LEAST 15 DAYS PRIOR TO ANY UTILITY DISRUPTIONS.

10. REFER TO INTERIOR PLUMBING DRAWINGS FOR TIE-IN OF ALL UTILITIES. ANY DISCREPANCIES MUST BE BROUGHT TO ENGINEER'S ATTENTION IMMEDIATELY.

11. THE UTILITY DRAWINGS DO NOT SHOW EVERY OFFSET, TRANSITION, AND FITTING THAT MAY BE REQUIRED. DETERMINE AND PROVIDE FITTINGS AND CONCRETE THRUST BLOCKS NECESSARY FOR COMPLETION OF THE SYSTEM.

12. PROVIDE EXTERIOR CORROSION PROTECTION ON UNDERGROUND METALLIC PIPE LINES UNLESS OTHERWISE NOTED.

13. ALL UNDERGROUND UTILITIES MUST BE INSTALLED A MINIMUM OF 5 FEET FROM NEW OR EXISTING STRUCTURES.

WATER NOTES:

1. A NCDEQ WATER EXTENSION PERMIT IS REQUIRED FOR THIS PROJECT.

2. ALL BACKFLOW PREVENTION ASSEMBLIES AND METERS MUST BE LOCATED WITHIN THE BUILDING MECHANICAL ROOMS UNLESS LOCATED OTHERWISE ON PLANS.

3. MINIMUM UTILITY SEPARATIONS

SANITARY SEWER

- 10- FEET (LATERALLY)
- 18-INCHES (VERTICAL - WATER OVER SEWER)

STORM SEWER

- 18-INCHES (VERTICAL)

4. WHERE A WATER MAIN PASSES OVER A SEWER, AND THE VERTICAL SEPARATION IS LESS THAN 18" OR WHERE WATER PASSES UNDER SEWER, THE SEWER AND WATERMAIN MUST BOTH BE DUCTILE IRON PIPE FOR A DISTANCE OF 10' ON EACH SIDE OF THE CROSSING (20' LENGTH CENTERED AT CROSSING - NO JOINTS)..

5. ALL WATERLINES MUST HAVE A MINIMUM OF 36-INCHES OF COVER FROM FINISHED GRADE, UNLESS DUCTILE IRON IS SPECIFIED.

6. ALL WATERLINES MUST HAVE A MINIMUM OF 18-INCHES OF SEPARATION BETWEEN UNDERGROUND POWER & COMMUNICATION LINES. COORDINATE WITH THE SITE ELECTRICAL / COMMUNICATIONS CONTRACTOR.

7. WHERE A WATERLINE PASSES UNDER A STORM DRAIN PIPE, THE WATERLINE MUST MAINTAIN A MINIMUM OF 18-INCHES OF VERTICAL SEPARATION FOR A DISTANCE OF 10' ON EACH SIDE OF THE CROSSING (20' LENGTH CENTERED AT CROSSING - NO JOINTS).

8. ALL WATERLINES MUST HAVE A MINIMUM OF 18-INCHES OF SEPARATION FROM GAS LINES. COORDINATE WITH SITE GAS CONTRACTOR.

SANITARY SEWER NOTES:

1. A NCDEQ SEWER EXTENSION PERMIT IS NOT REQUIRED FOR THIS PROJECT.

2. ALL CLEANOUT SYMBOLS SHOWN REPRESENT THE LOCATION OF SURFACE ACCESS POINTS. CONTRACTOR MUST LOCATE WYE APPROPRIATELY BASED ON PIPE SIZE AND PIPE DEPTH.

3. MAXIMUM DISTANCE BETWEEN CLEANOUTS ON SERVICE LINES IS 100- FEET. CLEANOUTS MUST BE PLACED AT VERTICAL AND HORIZONTAL CHANGES OF DIRECTION.

4. MANHOLES AND CLEANOUTS LOCATED IN PAVEMENT, CONCRETE OR OTHER TRAFFIC AREAS MUST BE SET FLUSH TO FINISHED GRADE AND BE HEAVY-DUTY TRAFFIC RATED (HS-20).

5. MANHOLES AND CLEANOUTS IN UNPAVED AND NON-TRAFFIC AREAS MUST BE SET 1-INCH ABOVE FINISHED GRADE.

6. ALL SEWER LINES MUST HAVE A MINIMUM OF 36-INCHES OF COVER FROM FINISHED GRADE UNLESS DUCTILE IRON PIPE.

7. MINIMUM UTILITY SEPARATIONS:

WATER

- SEE WATER NOTES

STORM SEWER

- 24-INCHES (VERTICAL)

8. ALL SEWER LINES MUST HAVE A MINIMUM OF 18-INCHES OF SEPARATION FROM UNDERGROUND POWER AND COMMUNICATION LINES. COORDINATE WITH SITE ELECTRICAL/COMMUNICATION CONTRACTOR.

9. ALL SEWER LINES MUST HAVE A MINIMUM OF 18-INCHES OF SEPARATION FROM GAS LINES.

EROSION CONTROL NOTES:

1. TOTAL DISTURBED AREA IS MORE THAN 1.0 ACRE. A LAND DISTURBANCE PERMIT IS REQUIRED.

2. ALL EROSION AND SEDIMENT CONTROL MEASURES MUST BE IN ACCORDANCE WITH THE NORTH CAROLINA SEDIMENTATION AND EROSION CONTROL DESIGN MANUAL (LATEST EDITION).

3. DIMENSIONS AND GRADES SHOWN ON THE PLANS MUST BE FIELD VERIFIED BY THE CONTRACTOR PRIOR TO CONSTRUCTION. NOTIFY THE ENGINEER IN WRITING IF DISCREPANCIES EXIST PRIOR TO PROCEEDING WITH CONSTRUCTION FOR NECESSARY PLAN OR GRADE CHANGES.

4. PROVIDE POSITIVE DRAINAGE AWAY FROM THE SITE AT ALL TIMES.

5. IF UNFORESEEN EROSION CONDITIONS DEVELOP DURING CONSTRUCTION AS THE RESULT OF INCREASED RUNOFF OR SEDIMENT DISPLACEMENT, THE CONTRACTOR SHALL TAKE ACTION TO REMEDY SUCH CONDITIONS AND PREVENT DAMAGE TO ADJACENT PROPERTIES.

6. EROSION AND SEDIMENT CONTROL MEASURES SHALL BE MAINTAINED CONTINUOUSLY, RELOCATED WHEN NECESSARY, AND SHALL BE CHECKED FOR STABILITY AND PROPER FUNCTIONING AFTER EVERY RAINFALL EVENT.

SEQUENCE OF LAND DISTURBANCE:

1. PRECONSTRUCTION CONFERENCE TO BE HELD PRIOR TO THE BEGINNING OF WORK.

2. DETERMINE AND MARK LIMITS OF DISTURBANCE.

3. INSTALL TEMPORARY CONSTRUCTION ENTRANCE.

4. INSTALL TEMPORARY SILT FENCE AND SILT FENCE ROCK OUTLETS.

5. TOPSOIL REMOVAL AND STORAGE. ALL ROOTS, STUMPS, OR OTHER DEBRIS SHALL BE REMOVED FROM TOP SOIL PRIOR TO STOCKPILING.

6. INSTALL PERMANENT DRAINAGE SWALES.

7. BRING SITE TO FINAL GRADE.

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8. INSPECT AND MAINTAIN ALL EROSION CONTROL DEVICES EVERY 7 DAYS AND AFTER EACH SIGNIFICANT RAINFALL EVENT (1.0" OR GREATER).

9. PLACE TEMPORARY SEEDING, INCLUDING MULCHING AND CRIMPING, IN ALL AREAS THAT ARE TO BE LEFT IDLE FOR MORE THAN 14 DAYS.

10. PERMANENT GRASS MUST BE INSTALLED PER PROJECT SPECIFICATIONS FOR ALL AREAS AT FINAL GRADE AND IN SEASON. FERTILIZE, WATER AND REPLACE TO ESTABLISH A VIGOROUS STAND OF GRASS. TEMPORARY IRRIGATION MAY BE REQUIRED UNTIL GRASS IS FULLY ESTABLISHED.

11. AFTER COMPLETION OF CONSTRUCTION AND THE SITE IS STABILIZED, REMOVE ALL ACCUMULATED SEDIMENT FROM SILT FENCE AND OTHER EROSION CONTROL MEASURES. GRASS THESE AREAS IMMEDIATELY AND REMOVE ALL EROSION CONTROL MEASURES.

12. CONTRACTOR IS RESPONSIBLE FOR SPREADING EXCAVATED SPOILS ON SITE AND GRADE WITH SMOOTH TRANSITIONS TO MATCH EXISTING GRADE. CONTRACTOR TO MAINTAIN POSITIVE DRAINAGE AT ALL AREAS. ALL IN ACCORDANCE WITH ALL LOCAL, STATE, AND FEDERAL REGULATIONS.

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12-18-2025

C-001

DEPARTMENT OF THE NAVY

NAVAL FACILITIES ENGINEERING COMMAND

MARINE CORPS BASE

CAMP LEJEUNE, NORTH CAROLINA

PRACTICAL APPLICATION TRAINING FACILITY (LE22S03R)—MARSOC

DES. WEB

DR. WEB

CHK. MKM

SUBMITTED BY: MKM

DESIGN DIR. CHRIS MAAS, PE

APPROVED: PWO OR OICC DATE

SATISFACTORY TO: DATE

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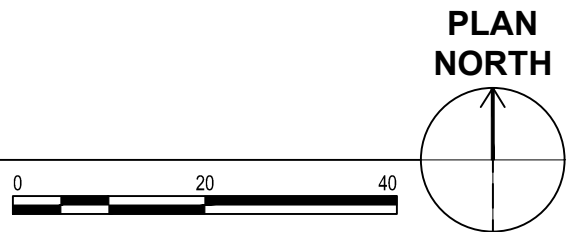
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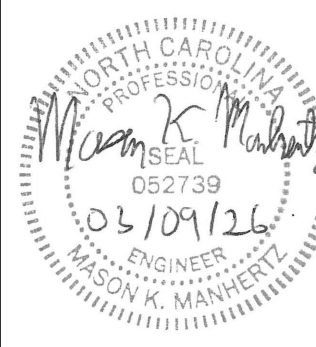
SHEET 08 OF 60



1 CIVIL DEMOLITION PLAN
1" = 20'-0"



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GENERAL SHEET NOTES

1. SEE SHEET C-001 AND C-002 FOR GENERAL NOTES, LEGEND, ABBREVIATIONS.

2. SEE SHEET VF100 FOR EXISTING CONDITIONS.

SHEET KEYNOTES

1. REMOVE EXISTING VEGETATION. NO CLEARING IS ALLOWED OUTSIDE OF AREA SHOWN.

2. REMOVE AND REPAIR GRAVEL PAVEMENT FOR WATER LINE AND HYDRANT INSTALLATION, MATCH EXISTING SECTION.

3. OPEN CUT/PATCH ASPHALT ROAD FOR UTILITY CONNECTION, SEE SHEET GU100.

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12-18-2025

CD100

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DR. WEB
CHK. MKM
SUBMITTED BY: MKM
DESIGN DIR. CHRIS MAAS, PE
APPROVED: PWO OR OICC

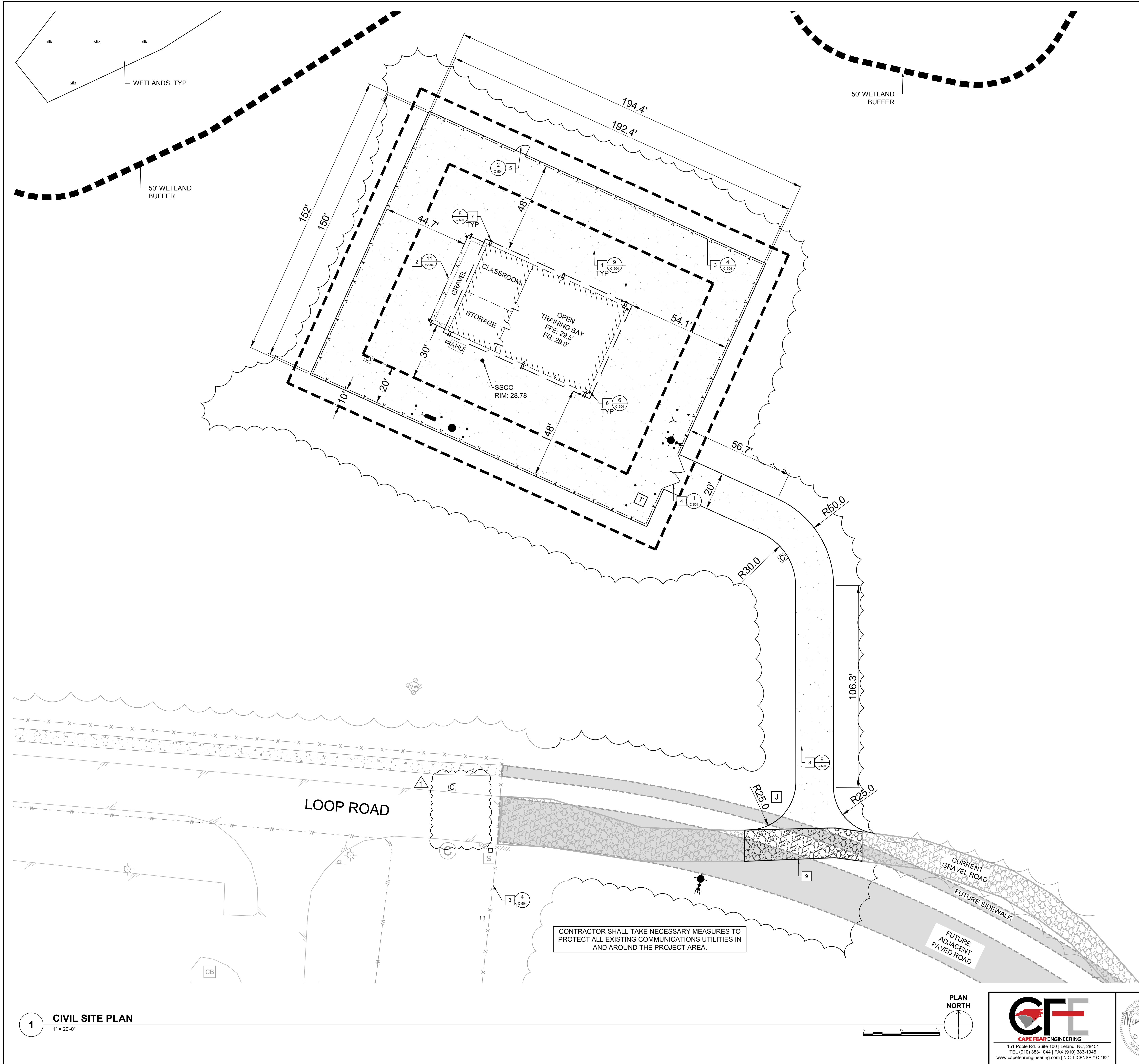
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DEPARTMENT OF THE NAVY
NAVAL FACILITIES ENGINEERING COMMAND
CAMP LEJEUNE, NORTH CAROLINA
PRACTICAL APPLICATION TRAINING
FACILITY (LE22S03R)-MARSOC
CIVIL DEMOLITION PLAN

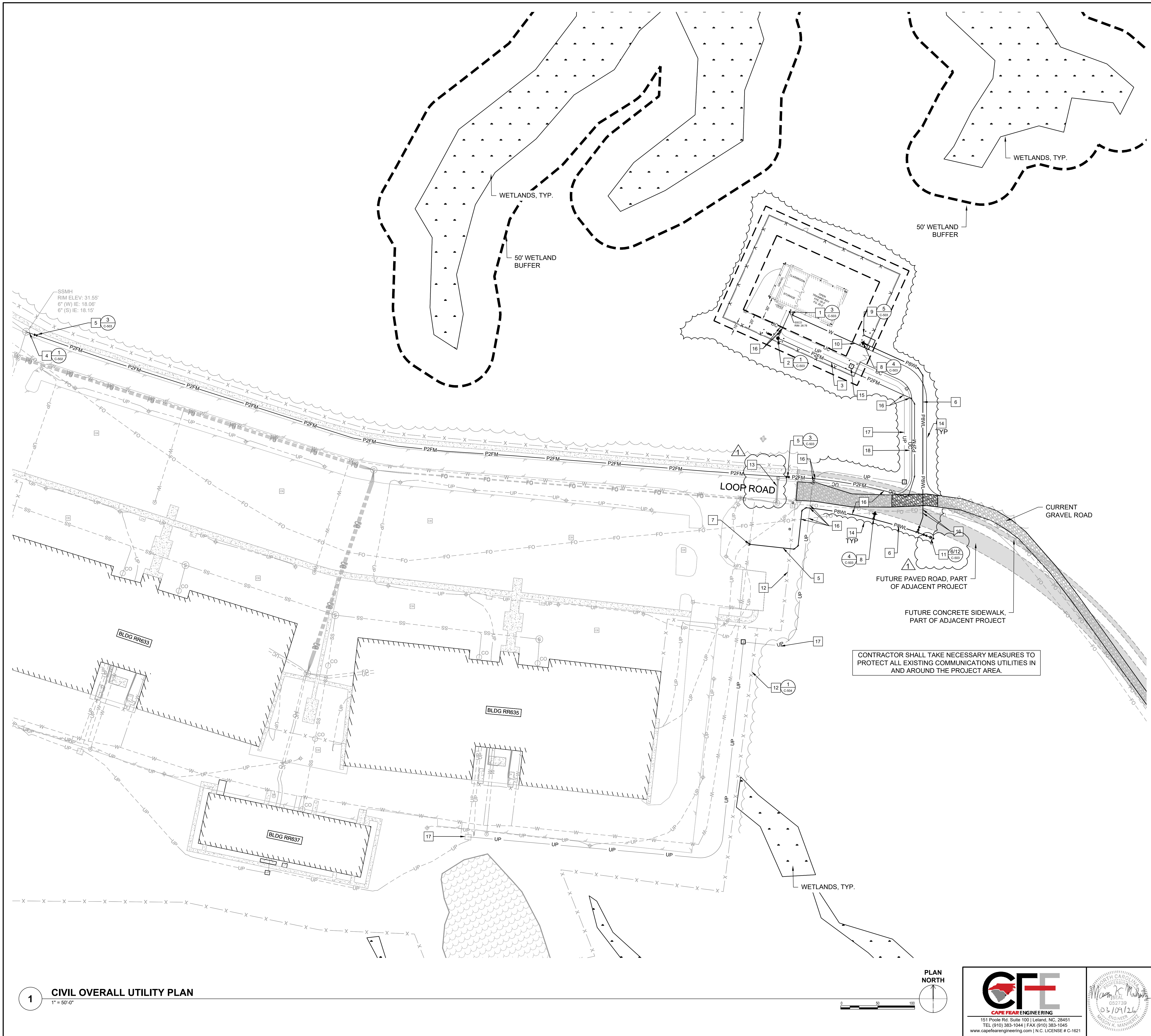
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GENERAL SHEET NOTES				
1. SEE SHEET C-001 AND C-002 FOR GENERAL NOTES, LEGEND, ABBREVIATIONS.				
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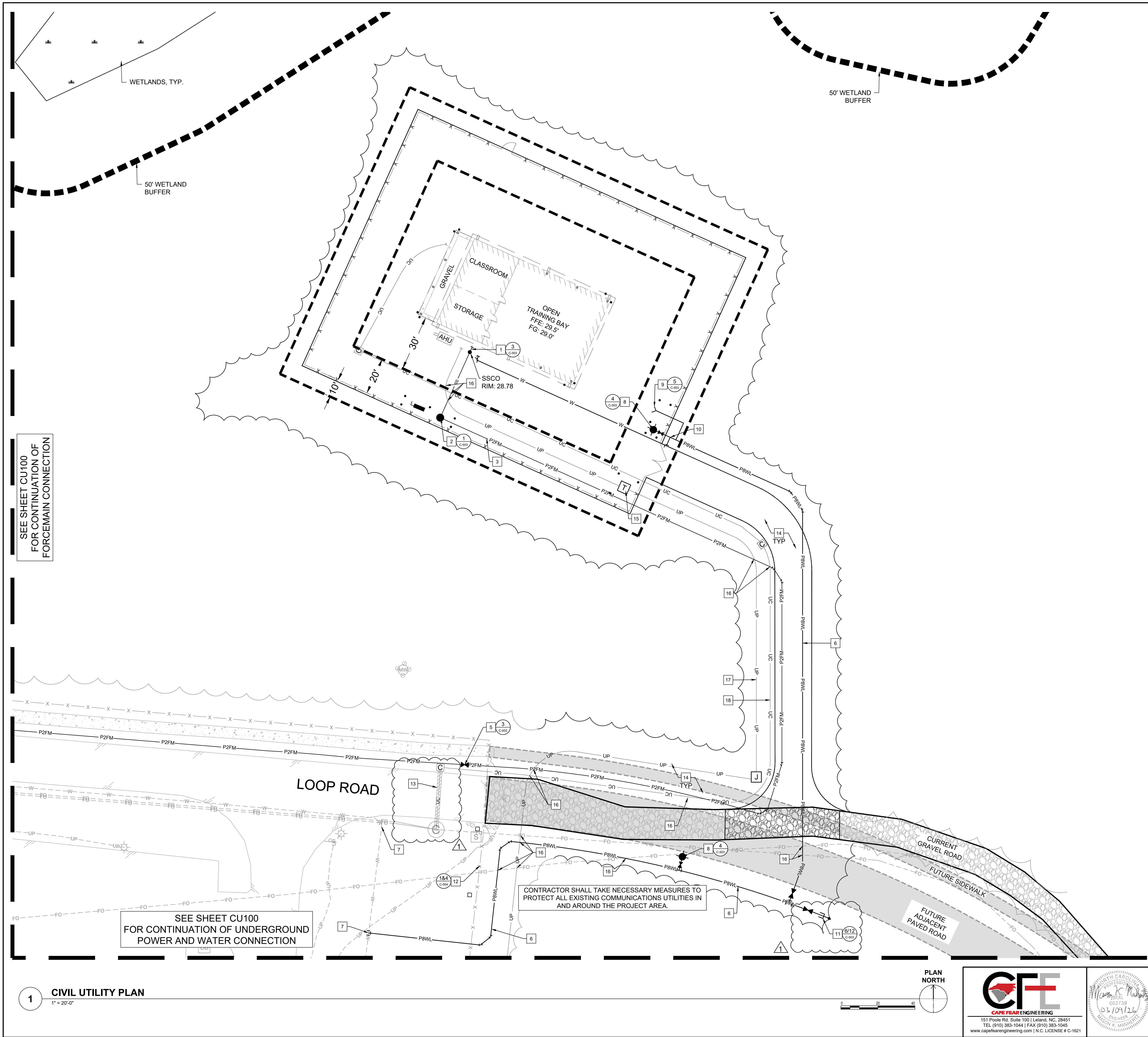
GENERAL SHEET NOTES

- SEE SHEET C-001 AND C-002 FOR GENERAL NOTES, LEGEND, ABBREVIATIONS.

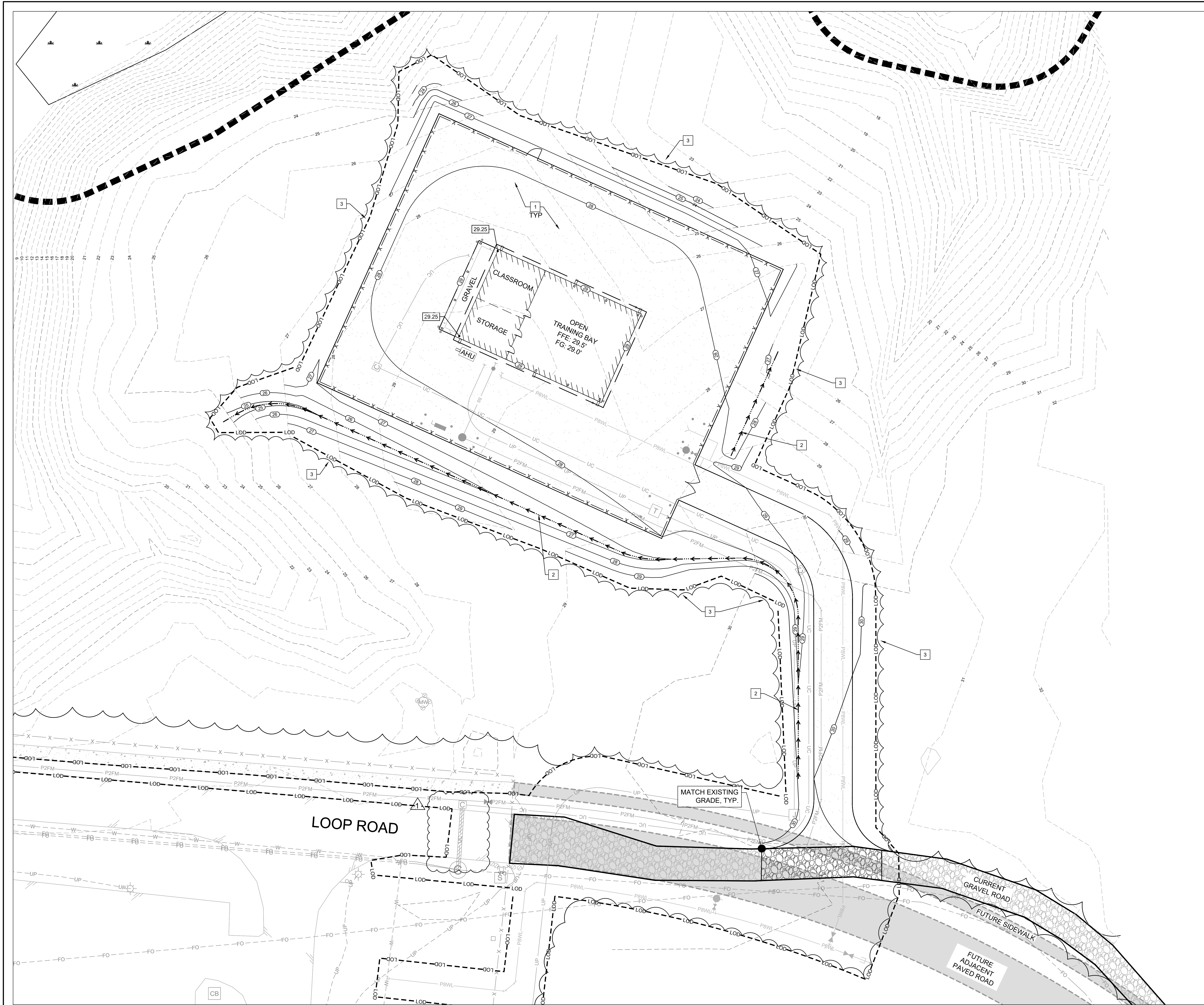
SHEET KEYNOTES

- 4" SEWER SERVICE WITH CLEANOUT (SCHEDULE 40 PVC).
- PACKAGE GRINDER PUMP STATION (SIMPLEX).
- 2" FORCEMAIN (SDR-21).
- CONNECT TO EXISTING SEWER STRUCTURE.
- 2" ISOLATION GATE VALVE.
- 8" WATER LINE C900 DR-18 (MINIMUM 36" DEPTH OF COVER).
- CONNECT TO EXISTING WATERLINE.
- FIRE HYDRANT.
- YARD HYDRANT.
- TAP FOR 1-1/2" DOMESTIC WATER SERVICE.
- 8" GATE VALVE AND TEMPORARY BLOW-OFF ASSEMBLY FOR FUTURE EXTENSION.
- PROTECT EXISTING FENCE TO REMAIN, EXCAVATE ON BOTH SIDES OF FENCE FOR UTILITY INSTALLATION.
- TRENCHLESS UTILITY INSTALLATION AT ROAD CROSSING (NO OPEN CUTTING). CONTRACTOR TO PROTECT EXISTING CONTROLLED GATE FEEDBACK LOOP.
- REMOVE TREES/VEGETATION AS REQUIRED FOR UTILITY INSTALLATION.
- PAD MOUNTED TRANSFORMER W/ PROTECTIVE BOLLARDS (SEE ELECTRICAL).
- UTILITY CROSSING, SEE DETAIL 8 ON SHEET C-503.
- UNDERGROUND ELECTRICAL LINE, CONNECT TO EXISTING TRANSFORMER (SEE ELECTRICAL).
- UNDERGROUND COMMUNICATIONS LINE (SEE ELECTRICAL SITE SHEET ES101).

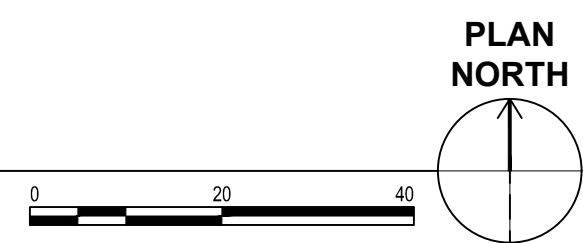
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DEPARTMENT OF THE NAVY NAVAL FACILITIES ENGINEERING COMMAND MARINE CORPS BASE CAMP LEJEUNE, NORTH CAROLINA			
PRACTICAL APPLICATION TRAINING FACILITY (LE22S03R)-MARSOC			
CIVIL OVERALL UTILITY PLAN			
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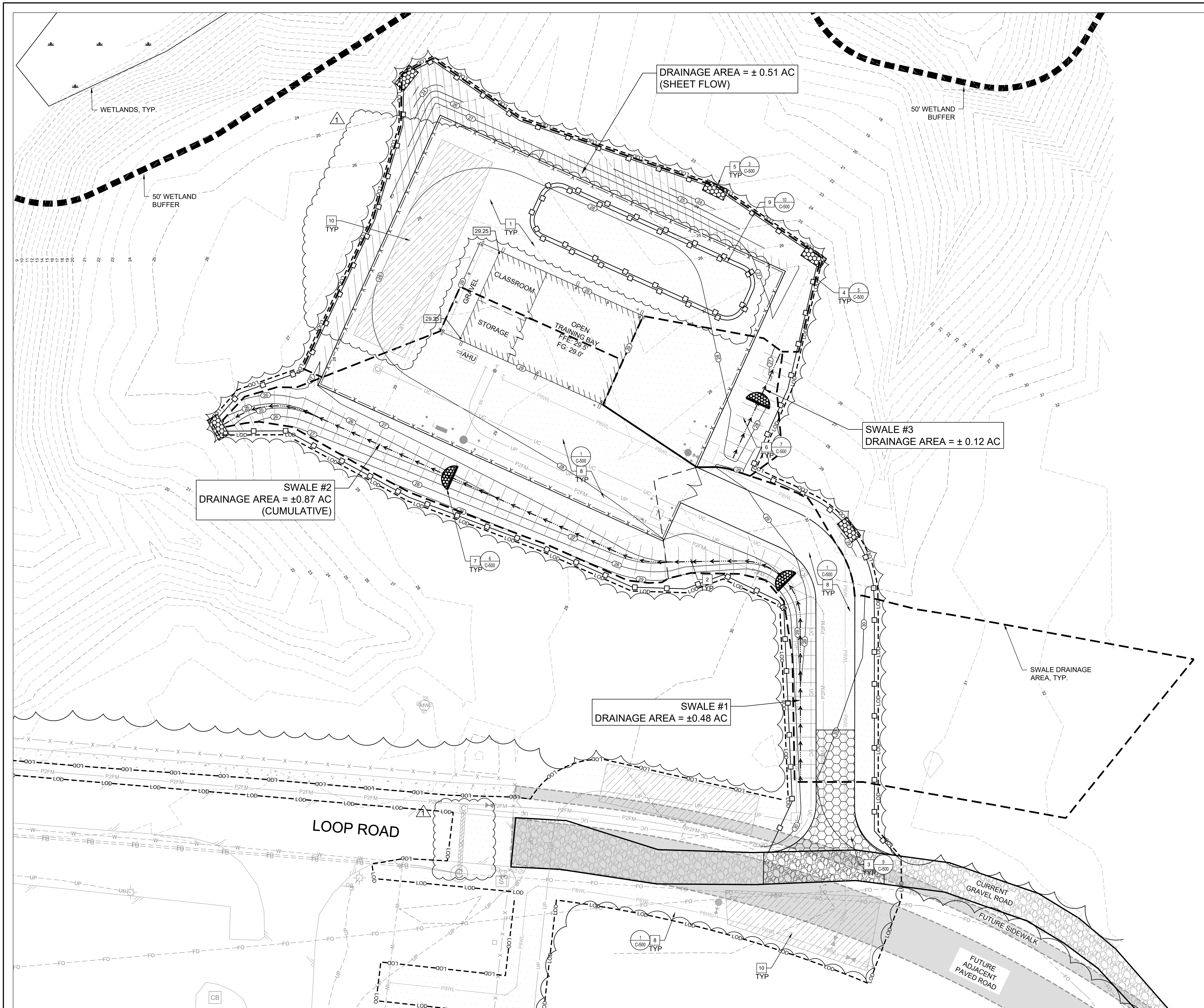
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GENERAL SHEET NOTES				
1. SEE SHEET C-001 AND C-002 FOR GENERAL NOTES, LEGEND, ABBREVIATIONS.				
SHEET KEYNOTES				
1. 4" SEWER SERVICE WITH CLEANOUT (SCHEDULE 40 PVC).				
2. PACKAGE GRINDER PUMP STATION (SIMPLEX).				
3. 2" FORCEMAIN (SDR-21).				
4. CONNECT TO EXISTING SEWER STRUCTURE (NOT USED - SEE SHEET CU100).				
5. 2" ISOLATION GATE VALVE.				
6. 8" WATER LINE C900 DR-18 (MINIMUM 36" DEPTH OF COVER).				
7. CONNECT TO EXISTING WATERLINE.				
8. FIRE HYDRANT.				
9. YARD HYDRANT.				
10. TAP FOR 1-1/2" DOMESTIC WATER SERVICE.				
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12. PROTECT EXISTING FENCE TO REMAIN, EXCAVATE ON BOTH SIDES OF FENCE FOR UTILITY INSTALLATION.				
13. TRENCHLESS UTILITY INSTALLATION AT ROAD CROSSING (NO OPEN CUTTING). CONTRACTOR TO PROTECT EXISTING CONTROLLED GATE FEEDBACK LOOP.				
14. REMOVE TREES/VEGETATION AS REQUIRED FOR UTILITY INSTALLATION.				
15. PAD MOUNTED TRANSFORMER W/ PROTECTIVE BOLLARDS (SEE ELECTRICAL).				
16. UTILITY CROSSING, SEE DETAIL 8 ON SHEET C-503.				
17. UNDERGROUND ELECTRICAL LINE, CONNECT TO EXISTING TRANSFORMER (SEE ELECTRICAL).				
18. UNDERGROUND COMMUNICATIONS LINE (SEE ELECTRICAL SITE SHEET ES101).				
FINAL			CU101	
12-18-2025				
DEPARTMENT OF THE NAVY			NAVAL FACILITIES ENGINEERING COMMAND	
MARINE CORPS BASE			CAMP LEJEUNE, NORTH CAROLINA	
PRACTICAL APPLICATION TRAINING FACILITY (LE22S03R)-MARSOC				
CIVIL UTILITY PLAN				
DES. WEB			SIZE	CODE IDENT. NO
DR. WEB			E1	80091
CHK. MKM			SCALE:	NOTED
SUBMITTED BY: MKM			SPEC.	05-22-0047
DESIGN DIR. CHRIS MAAS, PE			SHEET	13 OF 60
APPROVED: PWO OR OICC	DATE	NAVFAC DRAWING NO. 60037291A		
SATISFACTORY TO:	DATE	CONST. CONTR.		



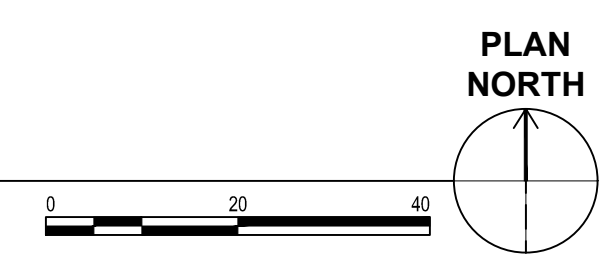
1 CIVIL GRADING PLAN
1" = 20'-0"



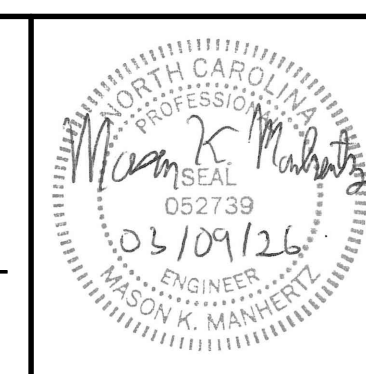
REVISIONS			
SYM		DATE	APPROVED
ASI	1	03/09/2026	
GENERAL SHEET NOTES			
1. SEE SHEET C-001 AND C-002 FOR GENERAL NOTES, LEGEND, ABBREVIATIONS. 2. CLIN IS REQUIRED FOR THIS PROJECT. SEE BIDDING INSTRUCTIONS AND DOCUMENTS, SHEET G-001, SPECIFICATIONS AND NOTES ON THIS SHEET. CLIN IS FOR A PER PLANT COST, INSTALLED. CLIN MAY BE EXERCISED AT OPTION OF GOVERNMENT. 3. LIST OF ITEMS WHICH MAY BE USED IN QUANTITY AND COMBINATIONS TO BE DETERMINED BY CONTRACTING OFFICER: A. UPRIGHT YAUPON HOLLY, 15 GALLON B. UPRIGHT YAUPON HOLLY, 30 GALLON C. EASTERN RED CEDAR, 15 GALLON D. EASTERN RED CEDAR, 30 GALLON E. WAX MYRTLE, 15 GALLON F. WAX MYRTLE, 30 GALLON 4. WHERE CLIN PLANTS ARE TO BE USED, PLANTING SPACING IS ON APPROXIMATELY 6' CENTERS			
SHEET KEYNOTES			
1. GRADE SITE TO PROVIDE POSITIVE DRAINAGE AWAY FROM BUILDING 2. DRAINAGE SWALE 3. APPROXIMATE LOCATIONS OF CLIN PLANTINGS, IF GOVERNMENT EXERCISES THE CLIN OPTION TO PROVIDE ADDITIONAL SCREENING FOR THE PROJECT. QUANTITY AND LOCATIONS AS DIRECTED BY CONTRACTING OFFICER. PLANTING LOCATIONS GENERALLY WILL FOLLOW LOD LINES, AND MAY BE INSIDE OR OUTSIDE THE LOD LINES.			
FINAL 12-18-2025		CG100	
		DEPARTMENT OF THE NAVY NAVAL FACILITIES ENGINEERING COMMAND MARINE CORPS BASE CAMP LEJEUNE, NORTH CAROLINA	
		PRACTICAL APPLICATION TRAINING FACILITY (LE22S03R)-MARSOC	
		CIVIL GRADING PLAN	
DES. WEB		SIZE E1	CODE IDENT. NO 80091
DR. WEB			NAVFAC DRAWING NO. 60037292
CHK. MKM			CONST. CONTR.
SUBMITTED BY: MKM			
DESIGN DIR. CHRIS MAAS, PE			
APPROVED: PWO OR OICC	DATE		
SATISFACTORY TO:	DATE		
SCALE: NOTED		SPEC. 05-22-0047	SHEET 14 OF 60



1 CIVIL EROSION CONTROL PLAN
1" = 20'-0"



CFE
CAPE FEAR ENGINEERING
151 Poole Rd. Suite 100 | Leland, NC, 28451
TEL (910) 383-1044 | FAX (910) 383-1045
www.capefearengineering.com | N.C. LICENSE # C-1621



REVISIONS		DATE	APPROVED
SYM	ASI 1	03/09/2026	

GENERAL SHEET NOTES

- SEE SHEET C-001 AND C-002 FOR GENERAL NOTES, LEGEND, ABBREVIATIONS.
- CONTRACTOR IS REQUIRED TO PAY THE ANNUAL NCG01 PERMIT FEE UNTIL THE CERTIFICATE OF COVERAGE IS TERMINATED.

SHEET KEYNOTES

- GRADE SITE TO PROVIDE POSITIVE DRAINAGE AWAY FROM BUILDING.
- LIMITS OF DISTURBANCE
- TEMPORARY CONSTRUCTION ENTRANCE
- TEMPORARY SILT FENCE
- TEMPORARY SILT FENCE ROCK OUTLET
- TEMPORARY CHANNEL LINER
- TEMPORARY STONE CHECK DAM
- PROVIDE GROUND COVER PER STABILIZATION SCHEDULE AS NECESSARY FOR ALL UTILITY TRENCHING AND LAND DISTURBANCE.
- APPROXIMATE STOCKPILE LOCATION. STOCKPILE SHOULD BE RELOCATED AS NECESSARY TO FACILITATE CONSTRUCTION SEQUENCING.
- APPROXIMATE CONTRACTOR LAYDOWN AREA. LAYDOWN AREA SHOULD BE RELOCATED AS NECESSARY TO FACILITATE CONSTRUCTION SEQUENCING.

REQUIRED GROUND COVER TIMES (SEE SEQUENCE OF LAND DISTURBANCE NOTES ON SHEET C-002 AND VEGETATION SCHEDULE DETAIL)

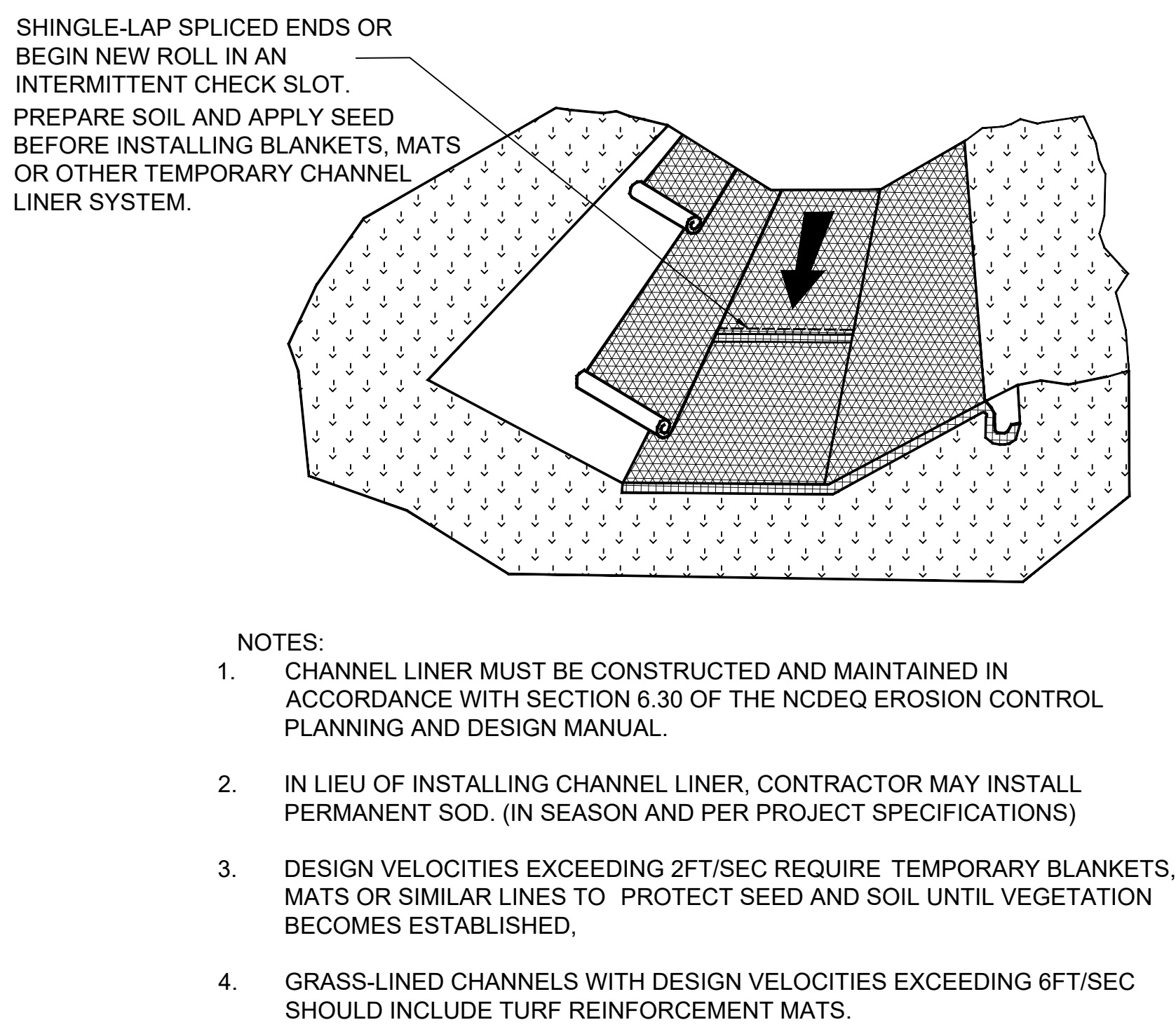
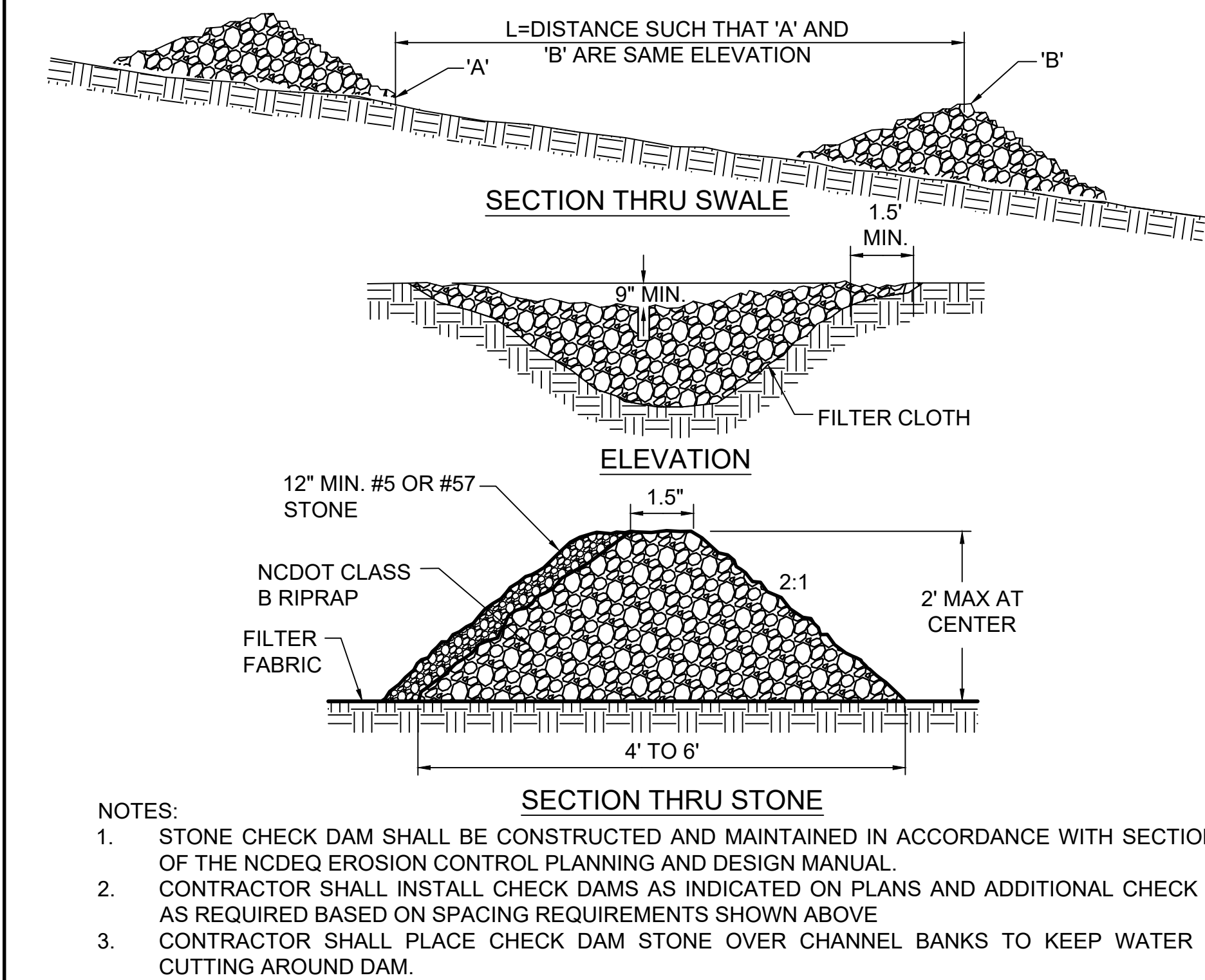
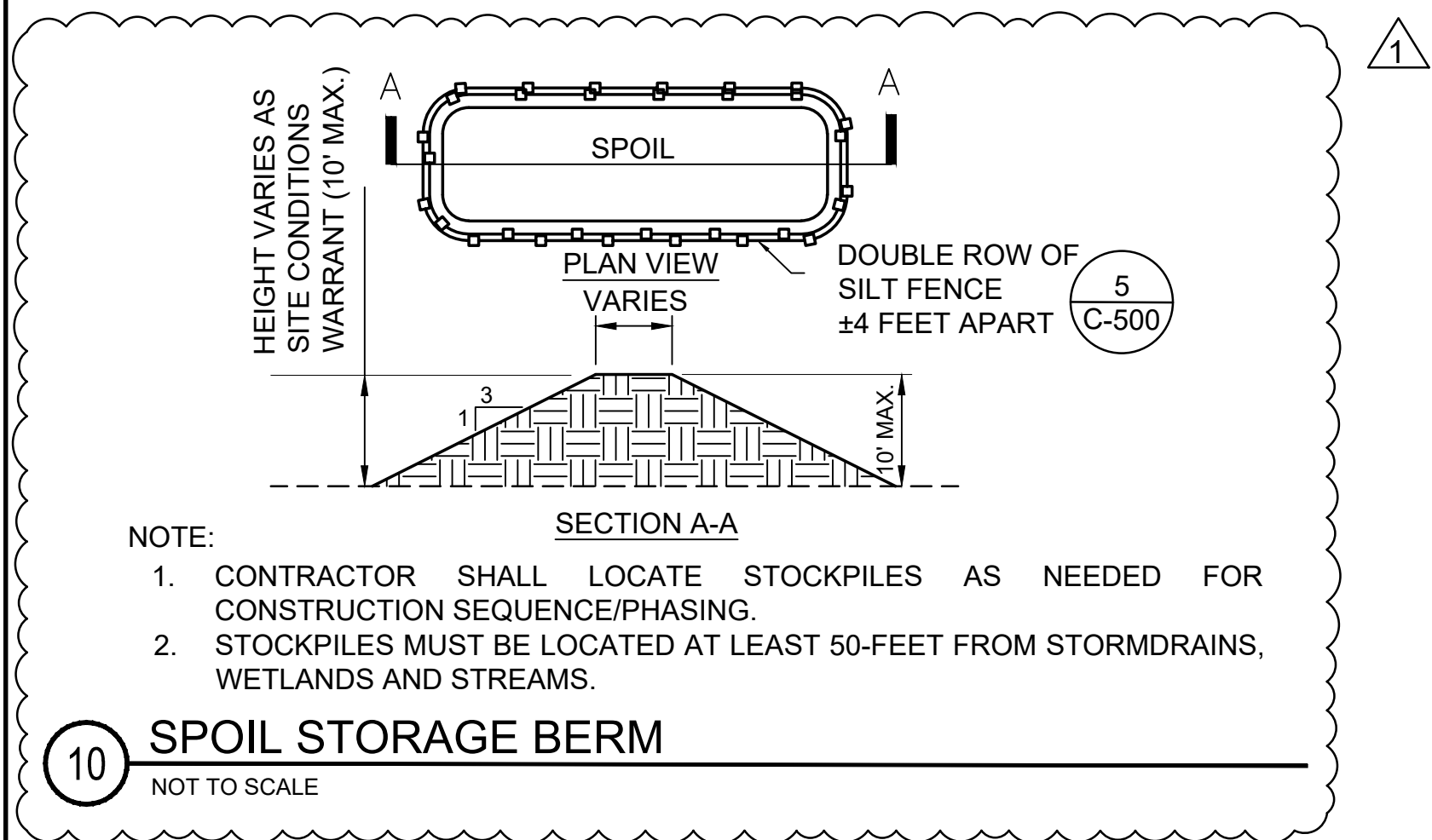
7 DAY MAXIMUM

14 DAY MAXIMUM

SPOIL STORAGE BERMS SHALL BE LOCATED BY THE CONTRACTOR AS NEEDED FOR CONSTRUCTION SEQUENCE/PHASING AND SHALL BE ENCLOSED IN A DOUBLE ROW OF TEMPORARY SILT FENCE.

FINAL 12-18-2025		CG101	
DEPARTMENT OF THE NAVY NAVAL FACILITIES ENGINEERING COMMAND MARINE CORPS BASE CAMP LEJEUNE, NORTH CAROLINA		PRACTICAL APPLICATION TRAINING FACILITY (LE22S03R)-MARSOC	
DES. WEB DR. WEB CHK. MKM SUBMITTED BY: MKM DESIGN DIR. CHRIS MAAS, PE APPROVED: PWO OR OICC DATE		CIVIL EROSION CONTROL PLAN	
SATISFACTORY TO: DATE		SIZE E1	CODE IDENT. NO 80091
		NAVAC DRAWING NO. 60037293	CONST. CONTR.
SCALE: NOTED		SPEC. 05-22-0047	SHEET 15 OF 60

REVISIONS			
SYM		DATE	APPROVED
△	ASI 1	03/09/2026	



MULCHING MATERIALS AND APPLICATION RATES

MATERIAL	RATE PER ACRE	QUALITY	NOTES
STRAW	2 TONS	DRY, UNCHOPPED, UNWEATHERED, AVOID WEEDS	SHOULD COME FROM WHEAT OR OATS SPREAD BY HAND OR MACHINE. MUST BE TACKED DOWN.

TEMPORARY SEEDING

TIME OF SEEDING	GRASS TYPE	AMOUNT/ ACRE	FERTILIZATION/ ACRE	FERTILIZATION/ACRE MAINTENANCE
DEC 1 - APR 15	RYE (GRAIN) & LESPEDEZA, KOBE	120 LBS. & 50 LBS.	750 LBS. 10-10-10	NA
APRIL 15-AUG 15	GERMAN MILLET	40 LBS.	750 LBS. 10-10-10	NA
AUG 15-DEC 30	RYE (GRAIN)	120 LBS.	1,000 LBS. 10-10-10	NA

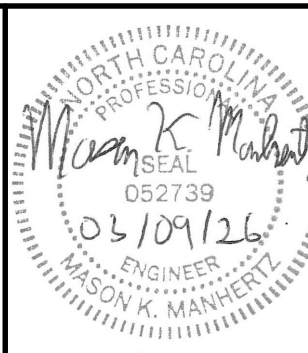
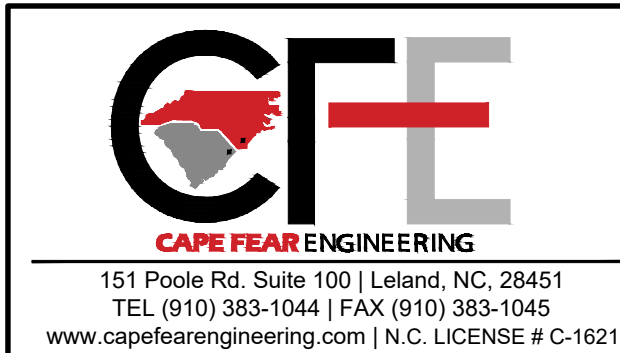
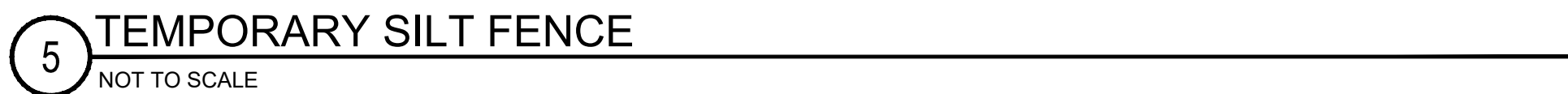
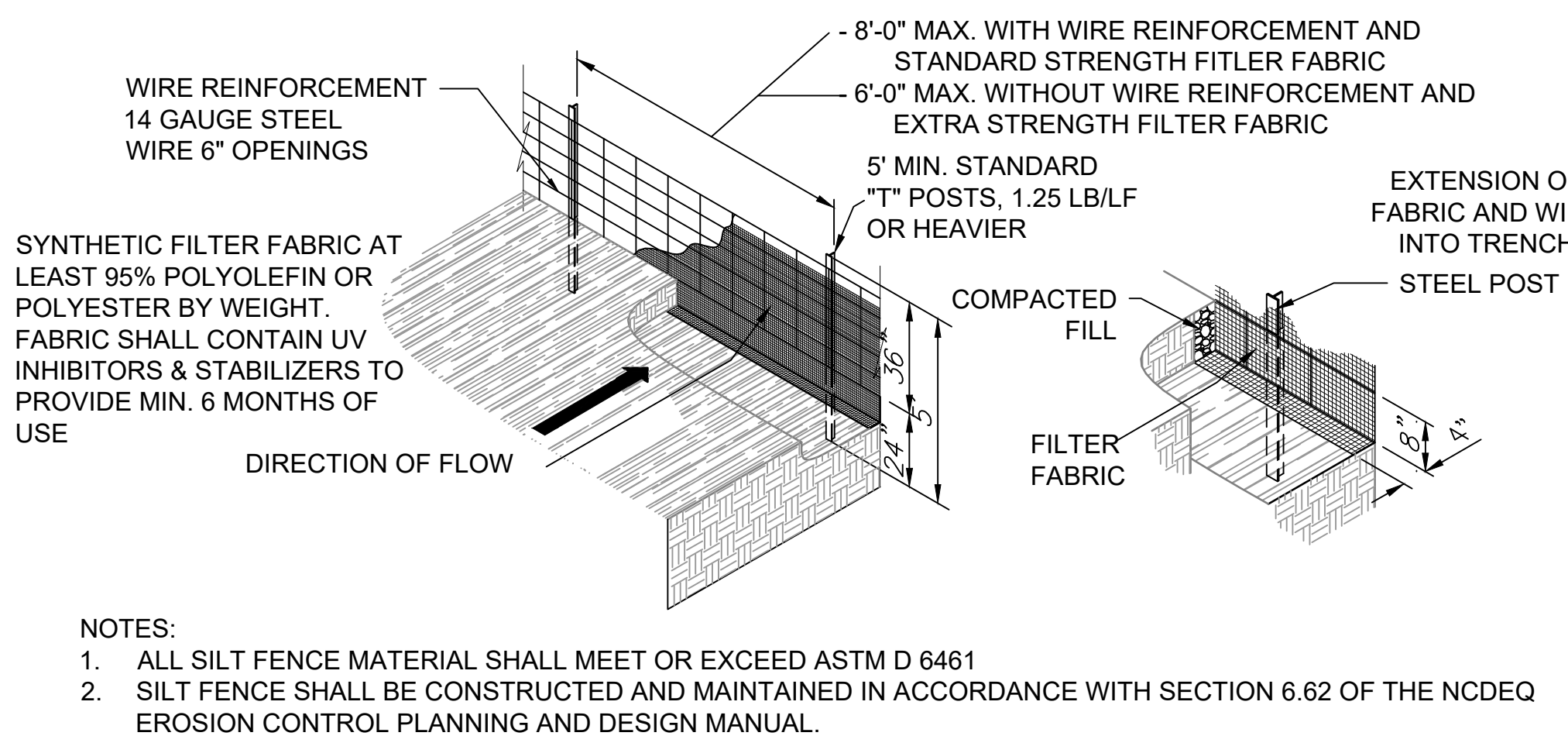
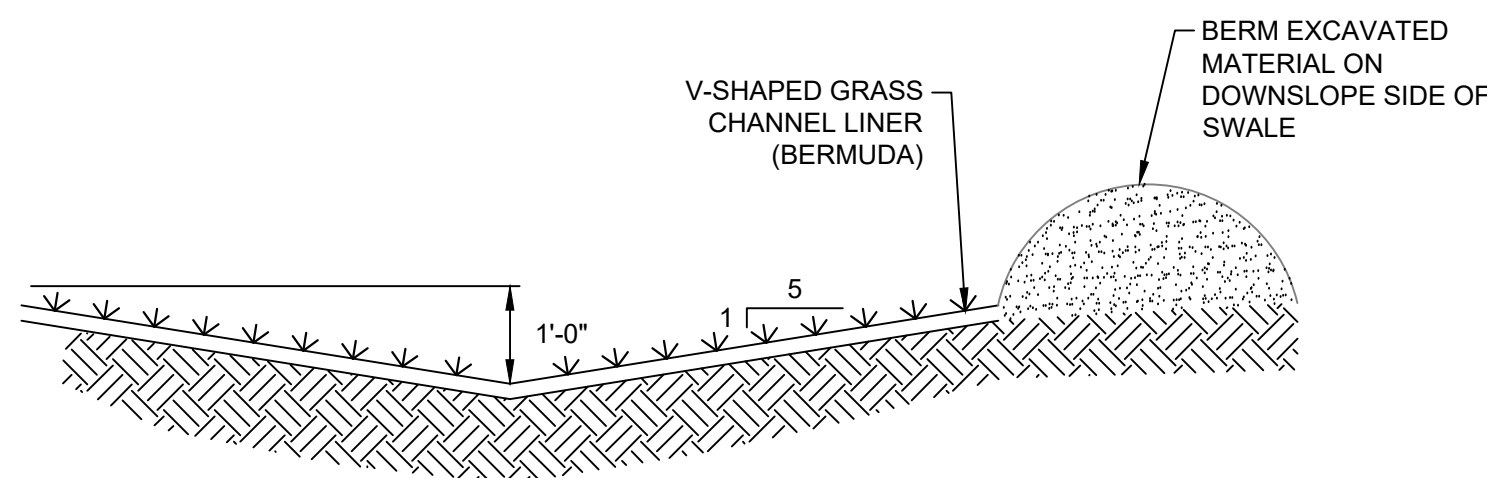
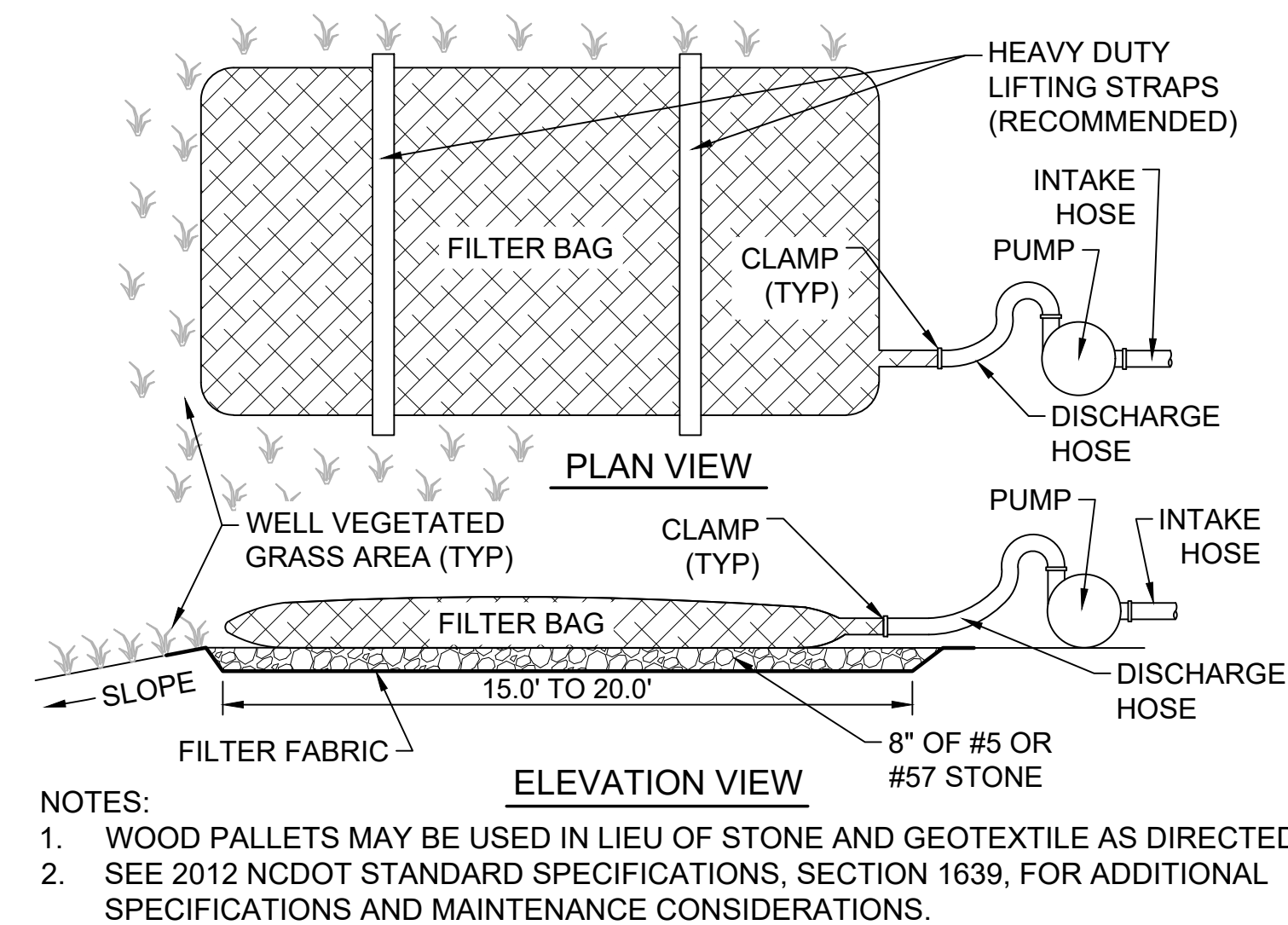
TEMPORARY STABILIZATION

TEMPORARY SEEDING SHALL BE INSTALLED AND MAINTAINED IN ACCORDANCE WITH SECTION 6.10 AND PERMANENT SEEDING SHALL BE INSTALLED AND MAINTAINED IN ACCORDANCE WITH SECTION 6.11 OF THE NCDEQ EROSION CONTROL PLANNING AND DESIGN MANUAL. SODDING SHALL BE INSTALLED AND MAINTAINED IN ACCORDANCE WITH SECTION 6.12 OF THE NCDEQ EROSION CONTROL PLANNING AND DESIGN MANUAL AND PROJECT SPECIFICATIONS.

PERMANENT STABILIZATION

PROVIDE SAND-BASED CENTIPEDE SOD (EREMOCHLOA OPHIUROIDES) FOR ALL TURF AREAS. SODDING SHALL BE INSTALLED AND MAINTAINED IN ACCORDANCE WITH SECTION 6.12 OF THE NCDEQ EROSION CONTROL PLANNING AND DESIGN MANUAL AND PROJECT SPECIFICATIONS. SEE SHEET C-001 FOR ADDITIONAL SPECIFICATIONS AND MAINTENANCE CONSIDERATIONS.

NEW STABILIZATION TIMEFRAMES		
SITE AREA DESCRIPTION	STABILIZATION	TIMEFRAME EXCEPTIONS
PERIMETER DIKES, SWALES, DITCHES AND SLOPES	7 DAYS	NONE
HIGH QUALITY WATER (HQW) ZONES	7 DAYS	NONE
SLOPES STEEPER THAN 3:1	7 DAYS	IF SLOPES ARE 10' OR LESS IN LENGTH AND ARE NOT STEEPER THAN 2:1, 14 DAYS ARE ALLOWED
SLOPES 3:1 OR FLATTER	14 DAYS	7 DAYS FOR SLOPES GREATER THAN 50' IN LENGTH.
ALL OTHER AREAS WITH SLOPES FLATTER THAN 4:1	14 DAYS	NONE, EXCEPT FOR PERIMETERS AND HQW ZONES

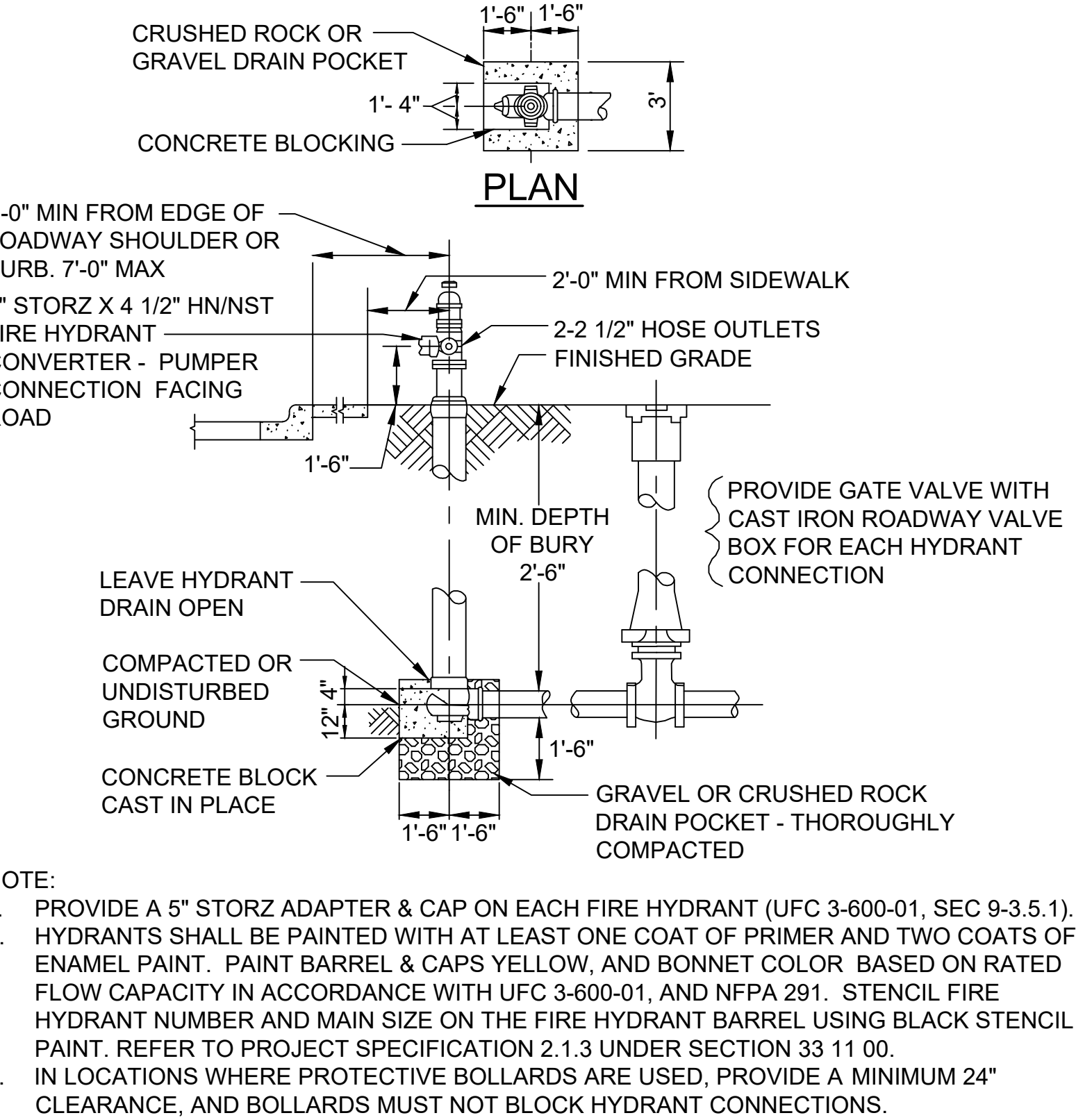


FINAL 12-18-2025			C-500			
			DEPARTMENT OF THE NAVY NAVAL FACILITIES ENGINEERING COMMAND			
			MARINE CORPS BASE CAMP LEJEUNE, NORTH CAROLINA			
			PRACTICAL APPLICATION TRAINING FACILITY (LE22S03R)-MARSOC			
			CIVIL EROSION DETAILS			
DES.	WEB		SIZE	CODE	IDENT. NO.	NAVFAC DRAWING NO. 60037294
DR.	WEB					
CHK.	MKM					
SUBMITTED BY: MKM			E1	80091	CONST. CONTR.	
DESIGN DIR. CHRIS MAAS, PE						
APPROVED: PWO OR OICC						
DATE			SCALE: NOTED			
SATISFACTORY TO:			SPEC. 05-22-0047		SHEET 16 OF 60	

REVISIONS			DATE	APPROVED
SYM				
Δ	ASI 1		03/09/2026	

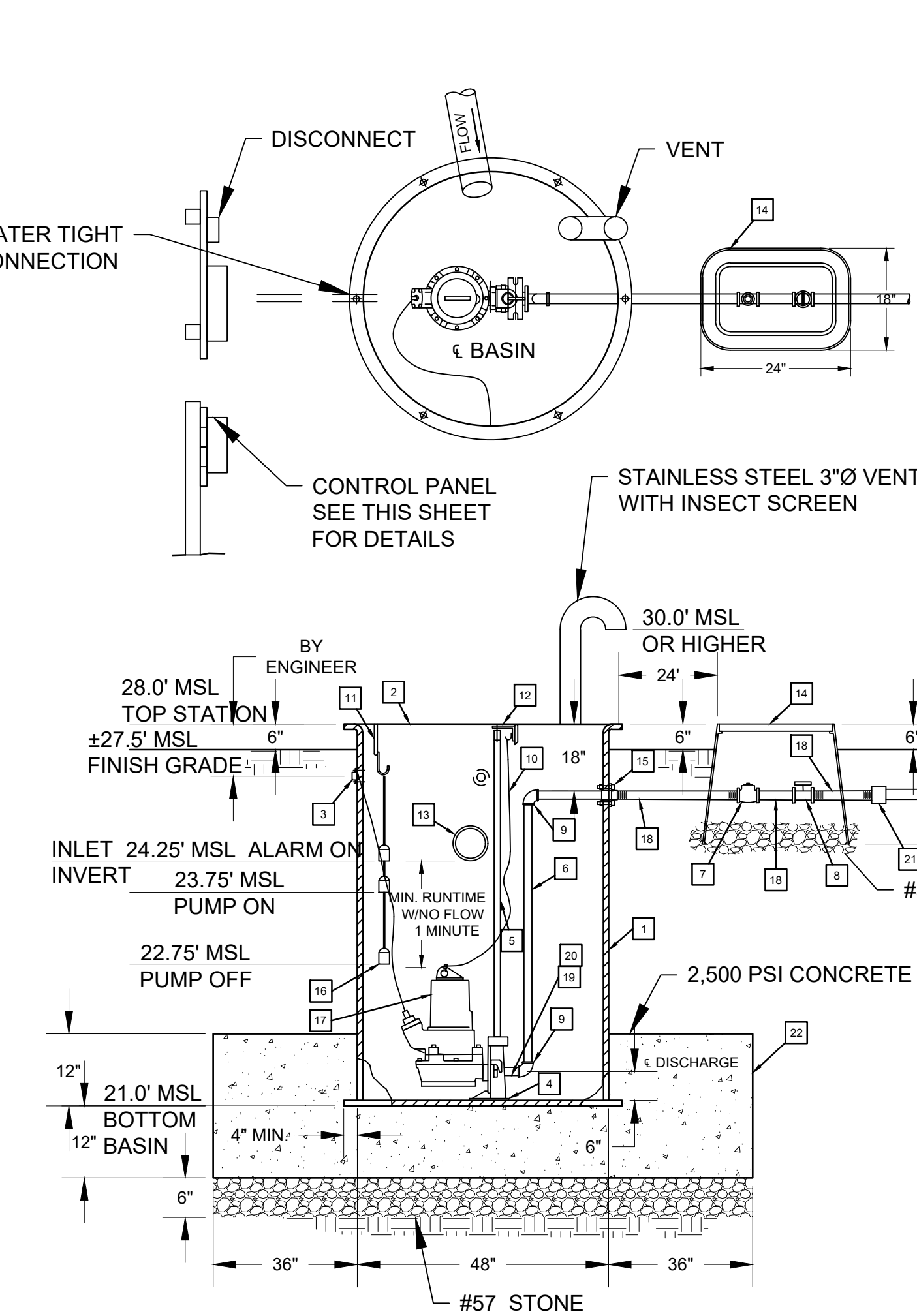
8 UTILITY CROSSING DETAIL

NOT TO SCALE



4 FIRE HYDRANT

NOT TO SCALE

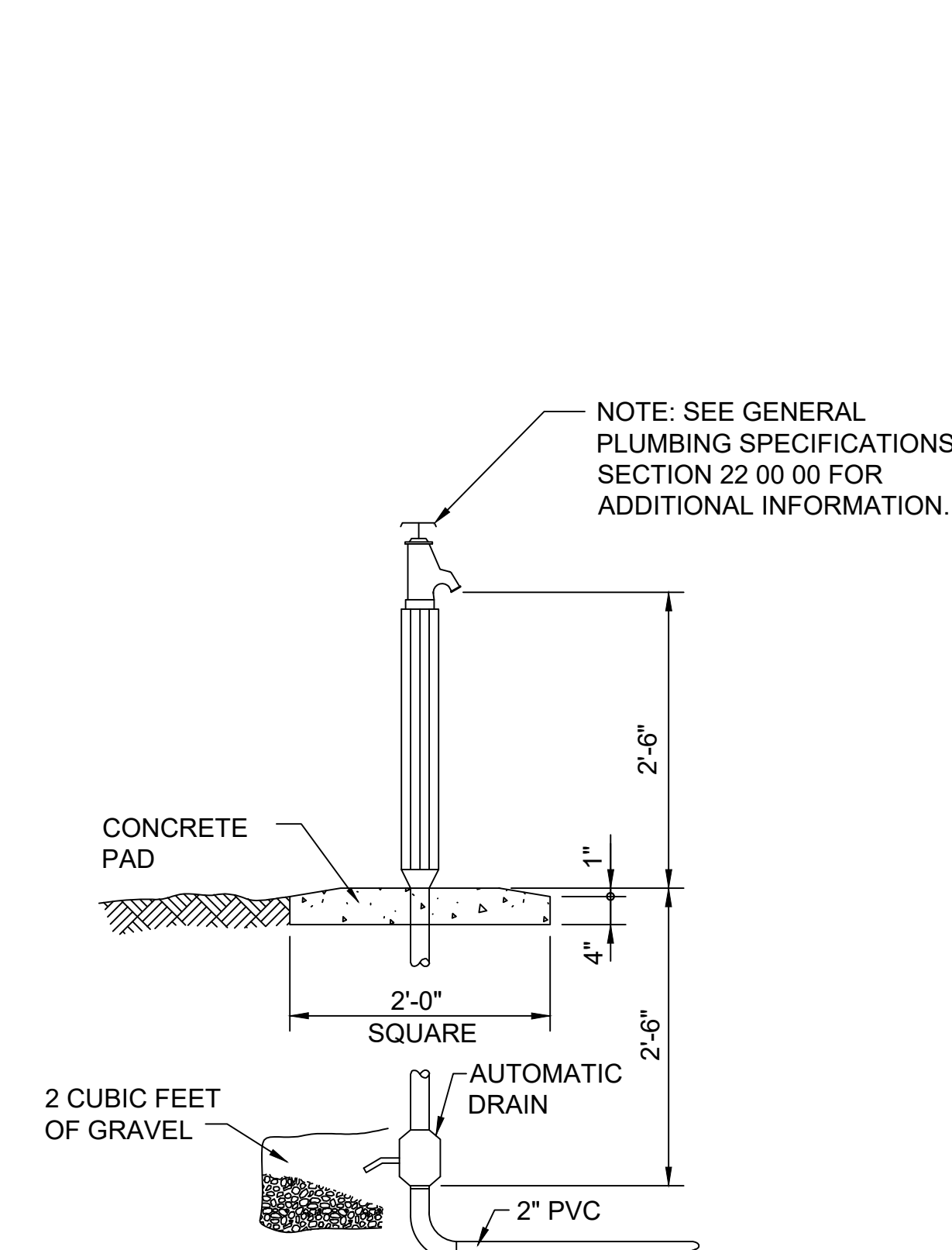


1 PACKAGED GRINDER PUMP STATION

NOT TO SCALE

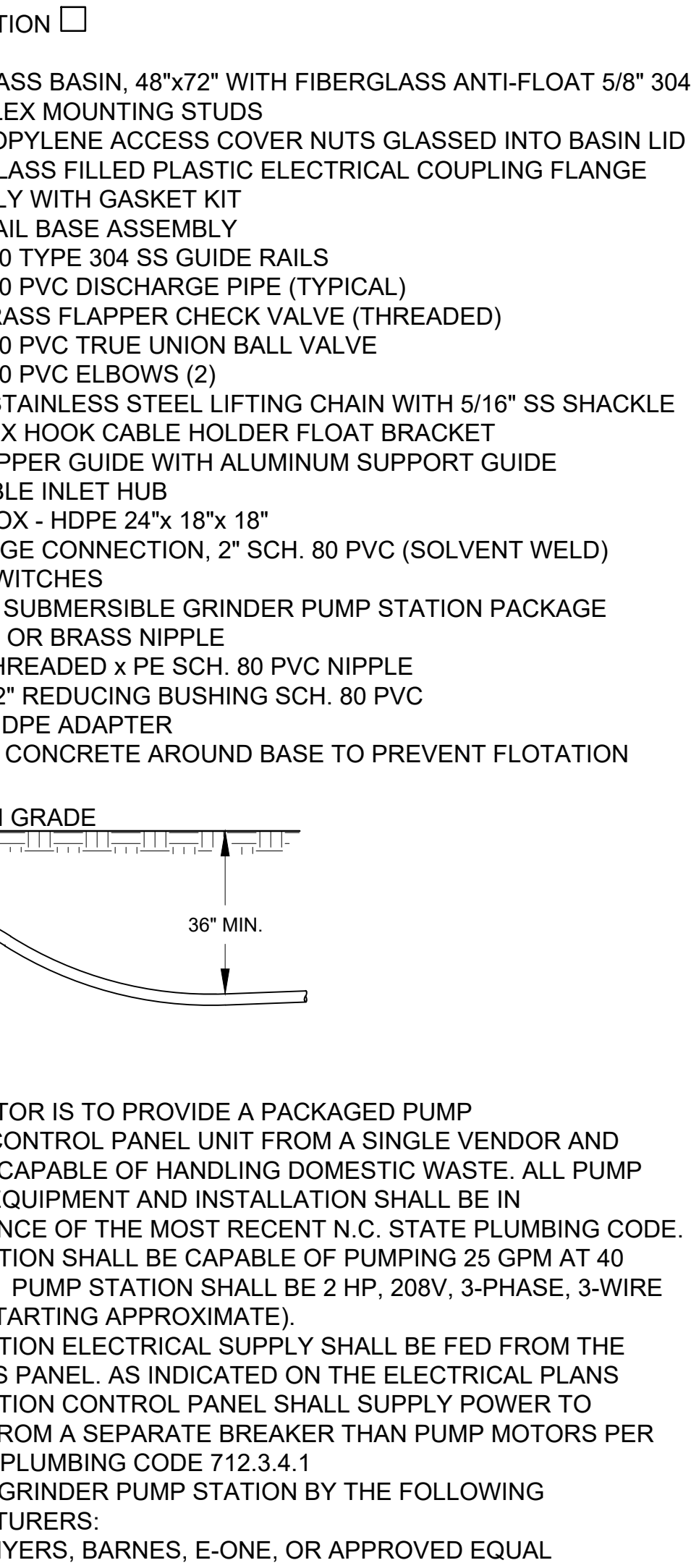
9 WATERLINE CONNECTION

NOT TO SCALE



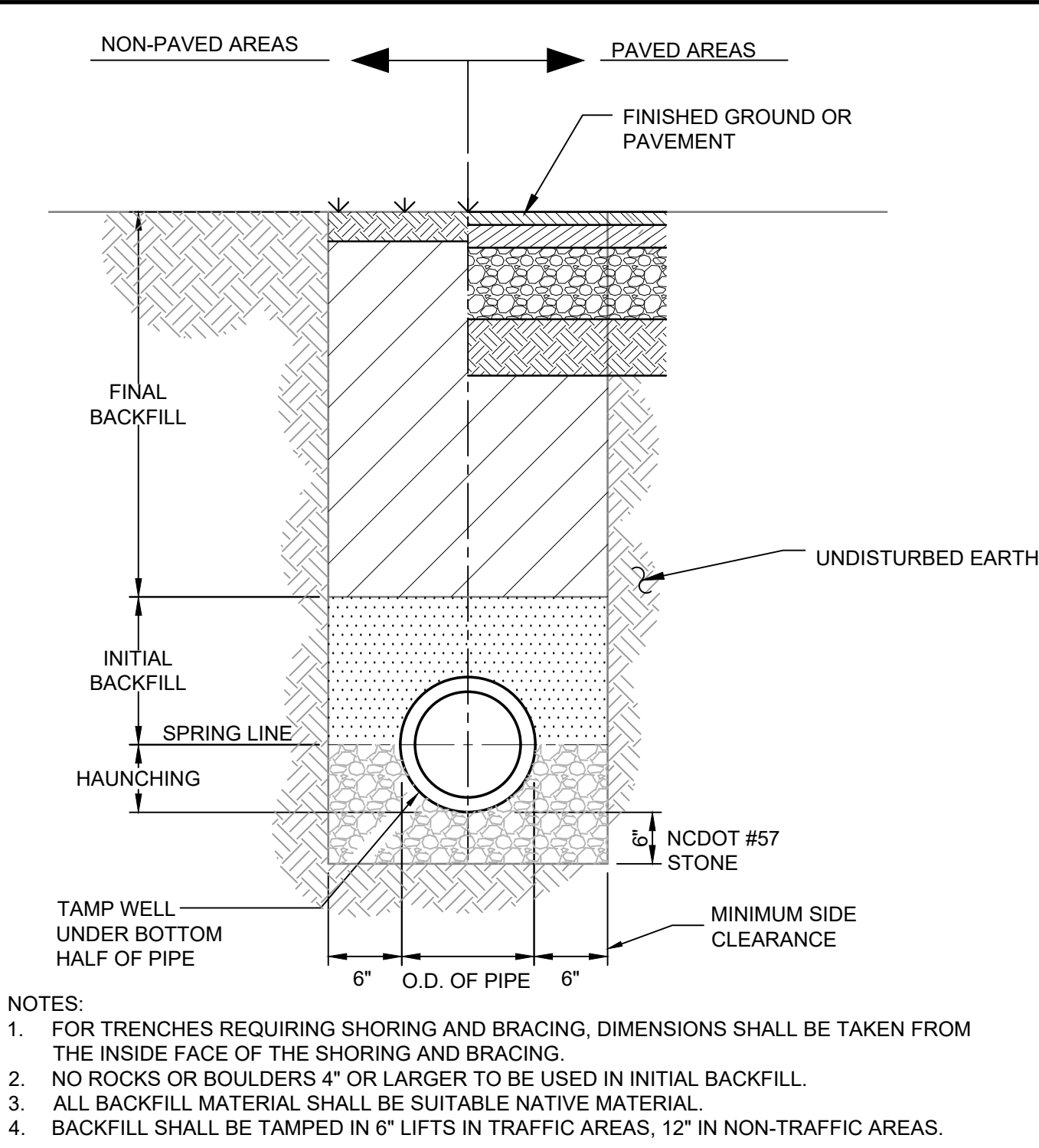
5 YARD HYDRANT

NOT TO SCALE



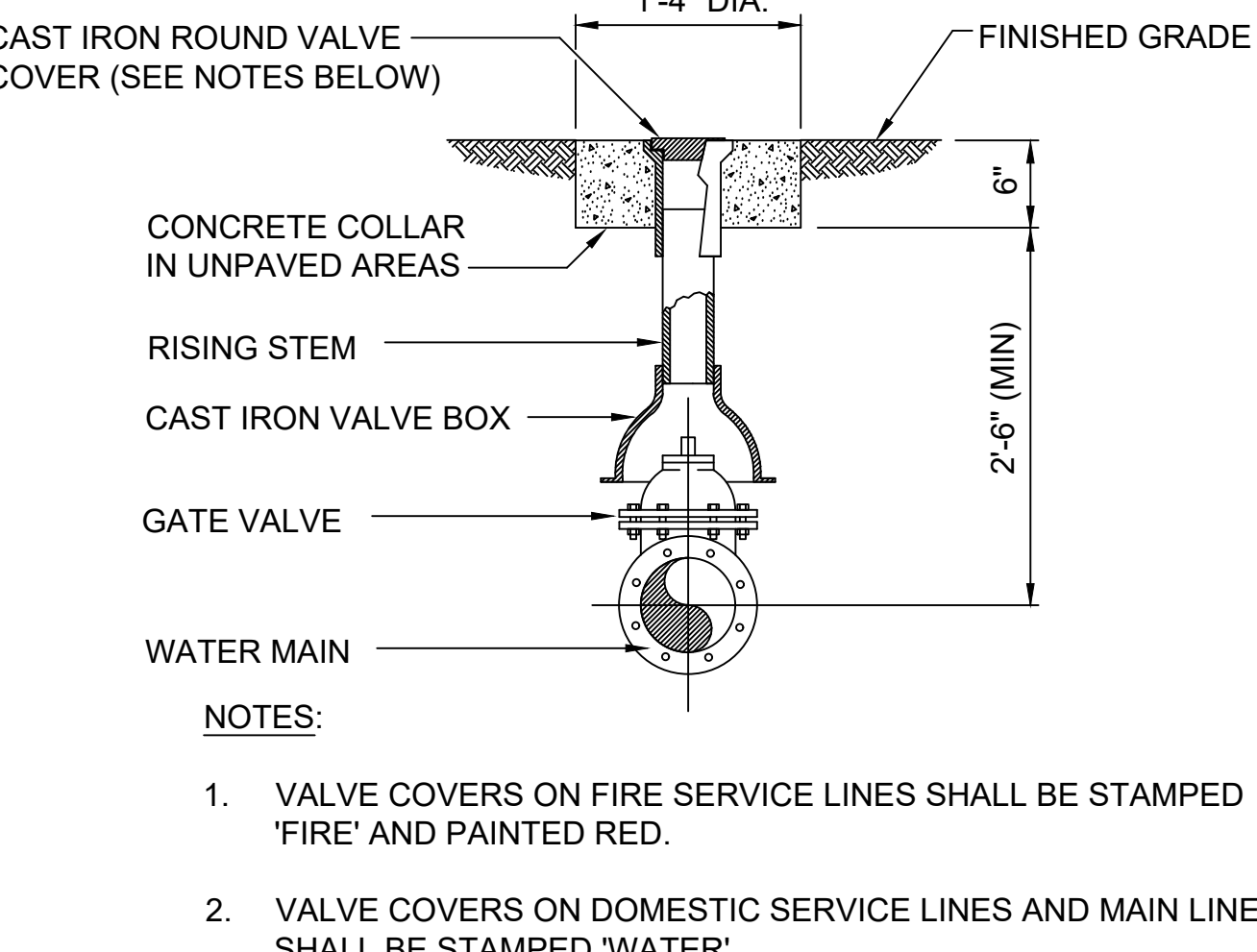
2 PUMP STATION CONTROL PANEL

NOT TO SCALE



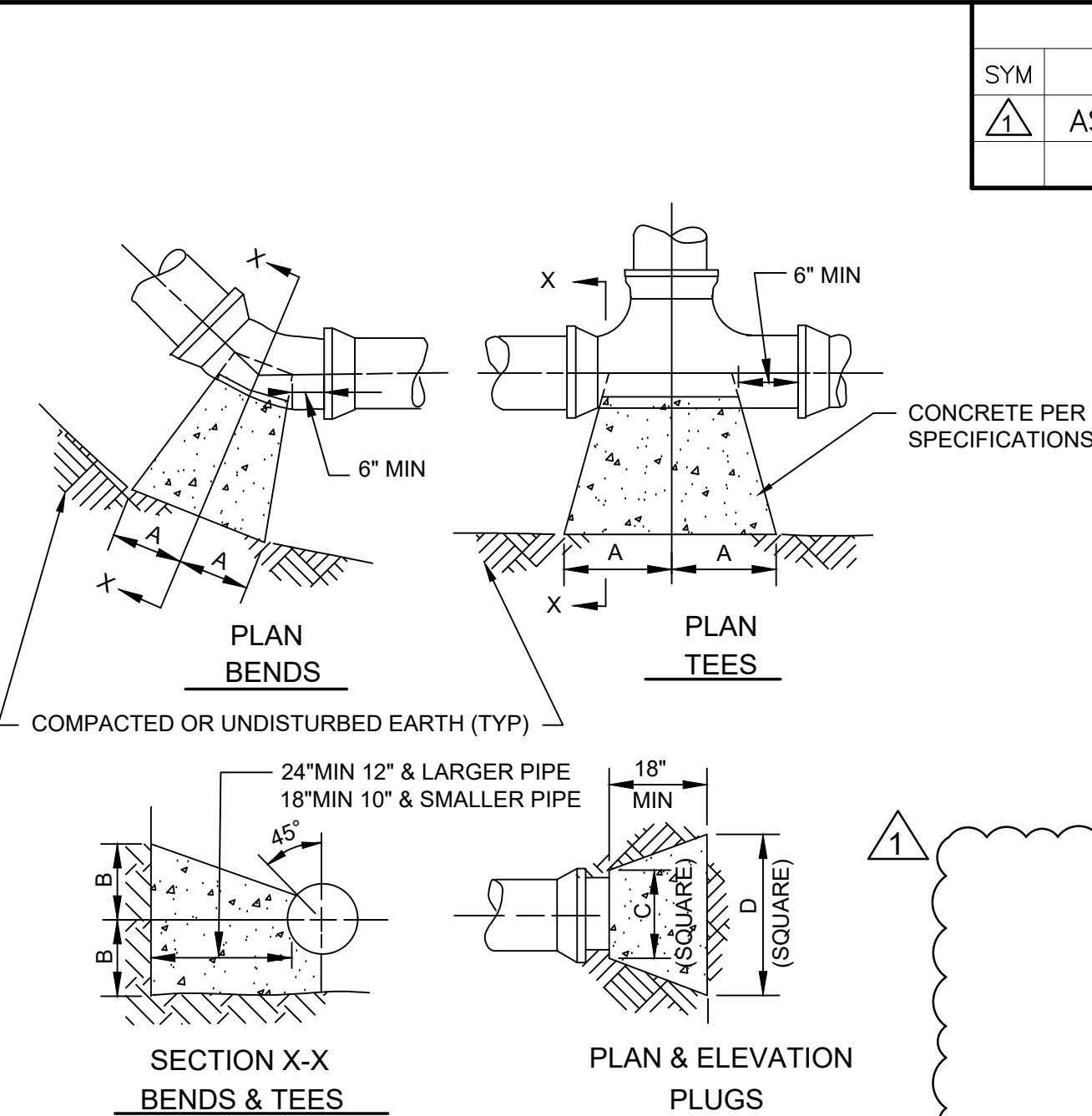
10 WATERLINE BEDDING DETAIL

NOT TO SCALE



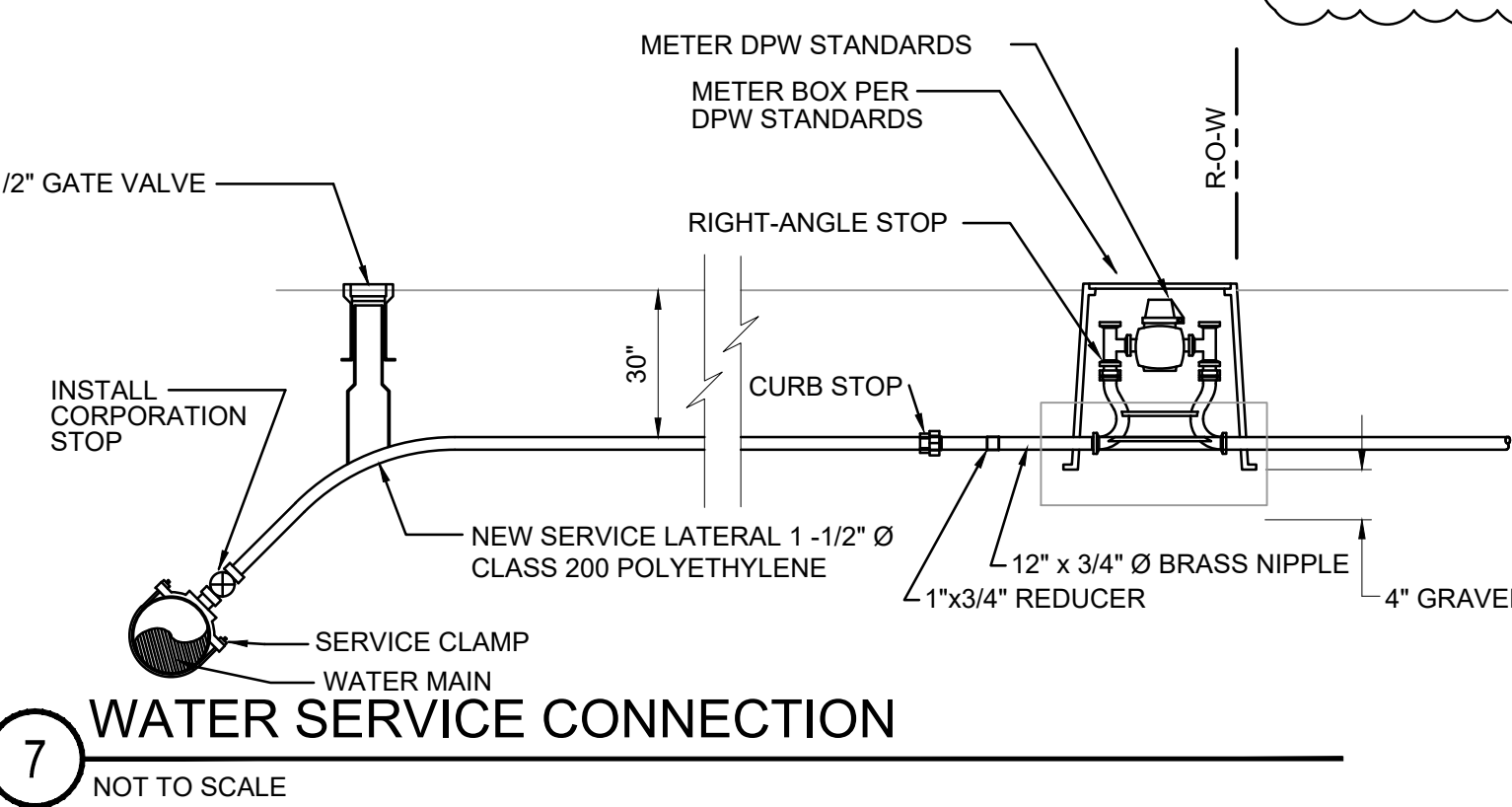
6 GATE VALVE

NOT TO SCALE



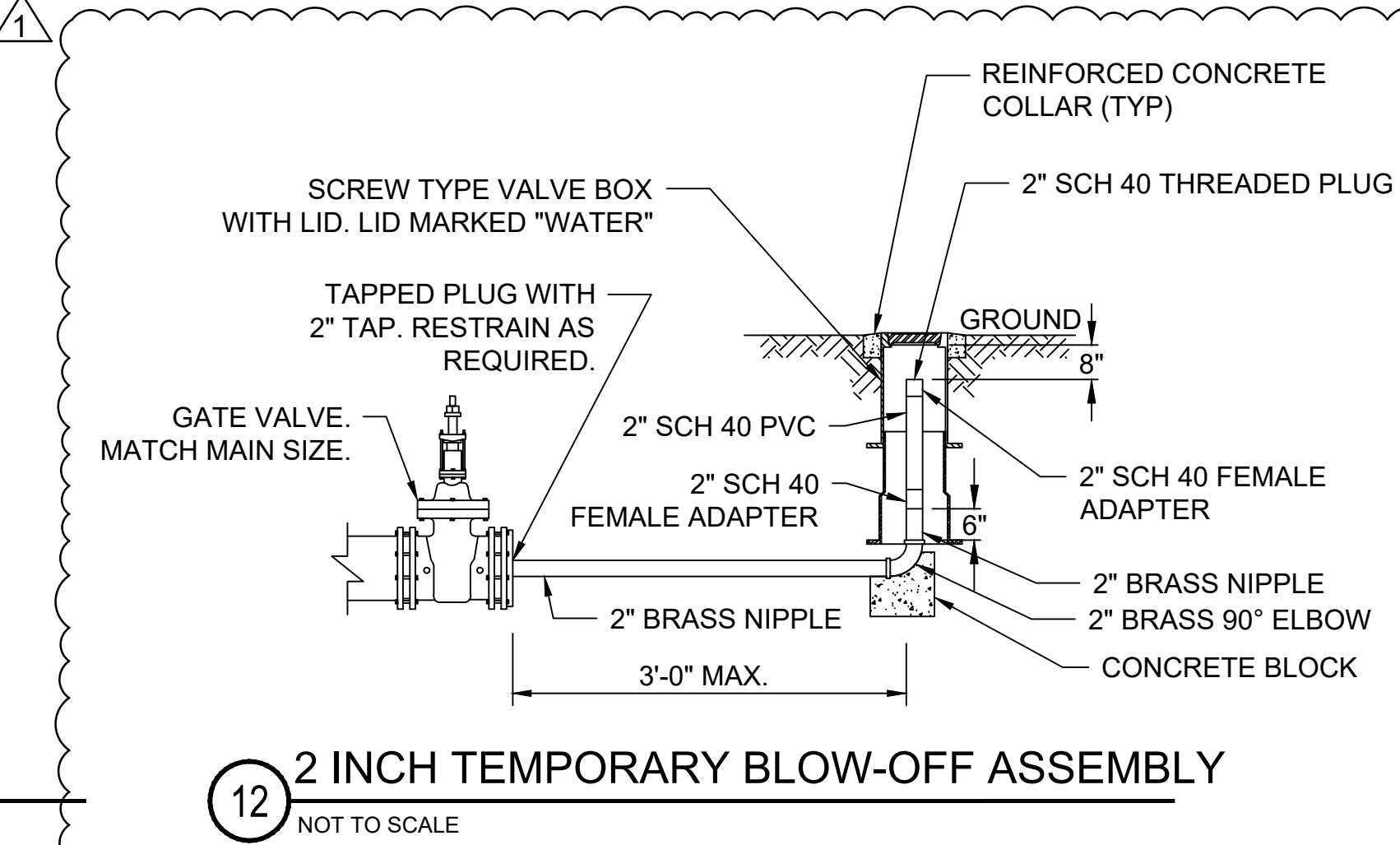
11 THRUST BLOCKS

NOT TO SCALE



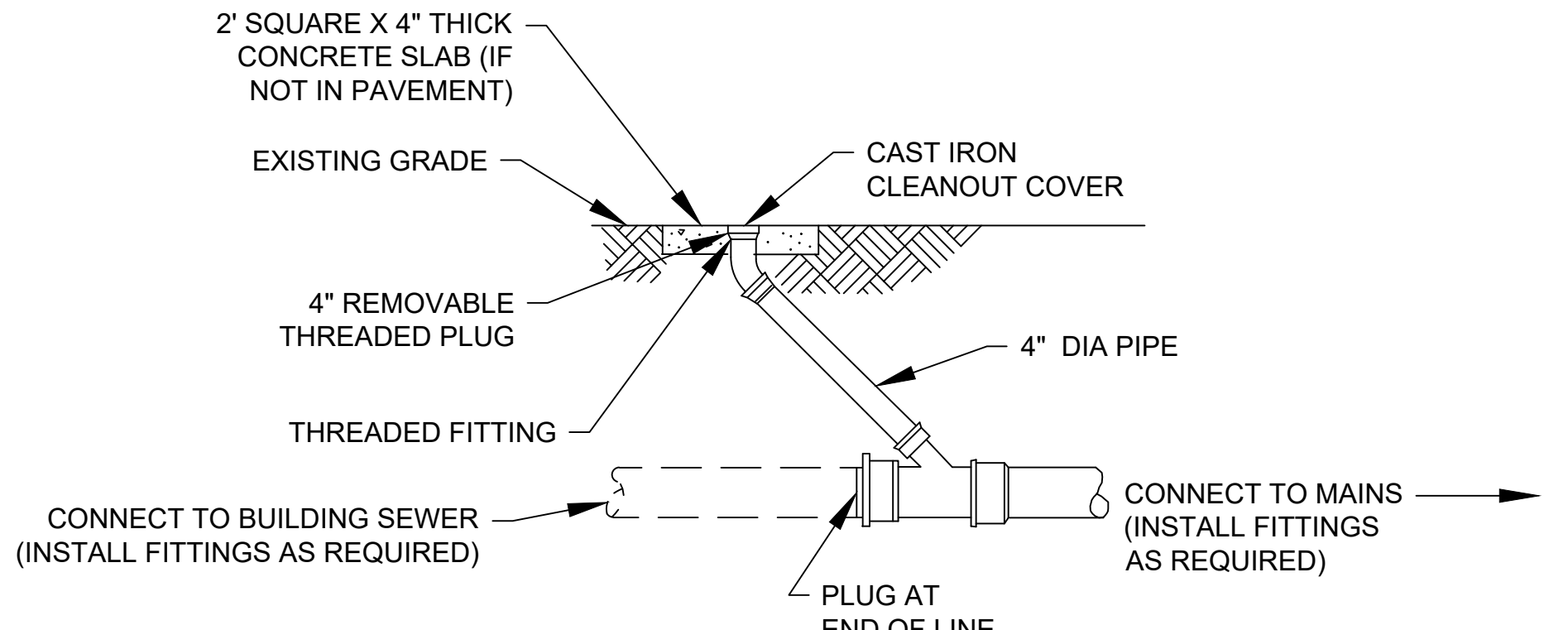
7 WATER SERVICE CONNECTION

NOT TO SCALE



12 2 INCH TEMPORARY BLOW-OFF ASSEMBLY

NOT TO SCALE



3 CLEANOUT/SERVICE

NOT TO SCALE

FINAL 12-18-2025		C-503	
DEPARTMENT OF THE NAVY		NAVAL FACILITIES ENGINEERING COMMAND	
MARINE CORPS BASE		CAMP LEJEUNE, NORTH CAROLINA	
DES. WEB		PRACTICAL APPLICATION TRAINING FACILITY (LE22S03R)–MARSOC	
DR. WEB			
CHK. MKM			
SUBMITTED BY: MKM		CIVIL UTILITY DETAILS	
DESIGN DIR. CHRIS MAAS, PE			
APPROVED: PWO OR OICC		DATE	
SATISFACTORY TO:		DATE	
SIZE	CODE IDENT. NO	NAVFAC DRAWING NO.	
E1	80091	60037297	
CONST. CONTR.			
SCALE: NOTED	SPEC. 05-22-0047	SHEET 19 OF 60	

